Merged Proofs of RMA

Contents

Problem 1

Solution to Problem 1: (Lack of) Quasi-Shift-Invariance
of the Φ^4_3 Measure

Statement and Answer

Let $\mathbb{T}^3$ be the unit 3-torus and μ the Φ^4_3 measure on $\mathcal{D}'(\mathbb{T}^3)$. For smooth $\psi \not\equiv 0$ let $T_\psi(u) = u + \psi$, and let $T^*_\psi\mu$ be the pushforward.

Theorem. *For every smooth $\psi \not\equiv 0$ and every admissible choice of mass and (nonzero) coupling constant, the measures μ and $T^*_\psi\mu$ are mutually* ***singular****.*

The answer is **NO**: μ and $T^*_\psi\mu$ are *not* equivalent. This is a simplified form of a result of Hairer–Peroni and rests on ideas of Hairer–Kusuoka–Nagoji.

Warning on a fatal false premise. It is tempting to write $d\mu/d\mu_0 = Z^{-1}e^{-\mathcal{R}(u)}$ against the Gaussian free field μ_0 and deduce quasi-invariance by a Cameron–Martin/Wick-binomial computation. This is *wrong in three dimensions*: Glimm showed that, unlike $d = 2$, the Φ^4_3 measure μ and the free field are mutually *singular* in $d = 3$ (the borderline dimension is $8/3$). There is no such Radon–Nikodym density, so any argument built on it collapses. The correct proof exhibits an event separating μ from $T^*_\psi\mu$, driven by a genuinely 3-dimensional logarithmic divergence.

Notation

Fix a space–time white noise ξ on $\mathbb{R} \times \mathbb{T}^3$ and let $\mathbf{l}$ be the stationary solution of $(\partial_t + 1 - \Delta)\mathbf{l} = \xi$. Write $P_N = P_{\le N}$ for the projection onto Fourier modes $|k| \le N$, $\mathbf{l}_N = P_N\mathbf{l}$, and $c_N = \mathbb{E}\mathbf{l}^2_N$. Let H_n be the Hermite polynomials normalized by $H_0 \equiv 1$, $H'_n = nH_{n-1}$, $\mathbb{E}H_n(Z, 1) = 0$ for standard normal Z. Set

$$\mathbf{q} = \lim_N H_2(\mathbf{l}_N, c_N), \qquad \mathbf{c} = \lim_N H_3(\mathbf{l}_N, c_N),$$

the Wick square and cube (convergent in $\mathcal{C}^{-1-2\kappa}$ and $\mathcal{C}^{-3/2-3\kappa}$ respectively), and let $\mathbf{C}$ solve $(\partial_t + 1 - \Delta)\mathbf{C} = \mathbf{c}$. We fix $0 < \kappa \le 1/100$. By [2], $\mathbf{c}$ and $\mathbf{C}$ are a.s. continuous in time with values in $\mathcal{C}^{-1/2-\kappa}$ and $\mathcal{C}^{1/2-3\kappa}$.

We will need the additional constant

$$c_{N,2} := \mathbb{E}\big[\mathbf{q}_N\, \mathbf{C}_N\big], \qquad \mathbf{q}_N := P_N\mathbf{q},\ \mathbf{C}_N := P_N\mathbf{C}. \tag{1}$$

The decomposition of μ

Proposition 1 ([4, 5, 6])**.** *There is a stationary process v, a.s. continuous in time with values in $\mathcal{C}^{1-2\kappa}(\mathbb{T}^3)$, such that*

$$u = \mathbf{l} - \mathbf{C} + v \tag{2}$$

is stationary with fixed-time law μ. Moreover v admits the paracontrolled form

$$v = -3\big((v - \mathbf{C}) \prec \mathbf{l}\big) + v^\sharp, \tag{3}$$

where $v^\sharp \in \mathcal{C}^{1+4\kappa}$ and $\prec$ is Bony's paraproduct.

The distinguishing event

For $\gamma > 0$ define

$$B_\gamma := \Big\{ u \in \mathcal{D}' : \lim_{N\to\infty} \big\langle (\log N)^{-\gamma} \big(H_3(P_N u; c_N) + 9 c_{N,2} P_N u \big), \psi \big\rangle_{\mathbb{T}^3} = 0 \Big\},$$

where N runs over $2^{\mathbb{N}}$. The next two lemmas (proved as in [2, Lems. 3.11–3.12]) supply the analytic input.

Lemma 1. *For $\gamma > \frac{1}{2}$ and each fixed $t > 0$, $(\log N)^{-\gamma} :\mathbf{l}_N^3:(t) \to 0$ a.s. in $\mathcal{C}^{-3/2}(\mathbb{T}^3)$ and in $L^p(\Omega; \mathcal{C}^{-3/2})$ for all $p > 0$.*

Sketch. With $\beta = -d/2 = -3/2$ and the embedding $W^{\beta,2p} \hookrightarrow \mathcal{C}^{\beta - d/2p}$, one bounds $\mathbb{E}\|(\log N)^{-\gamma} :\mathbf{l}_N^3:\|_{W^{-3/2,2p}}^{2p}$ by a constant times

$$(\log N)^{-2\gamma} \sum_{|\omega_i| \le N} \langle \omega_1 + \omega_2 + \omega_3 \rangle^{-3} \prod_i \langle \omega_i \rangle^{-2} \lesssim (\log N)^{-2\gamma+1},$$

which is summable along $N \in 2^{\mathbb{N}}$ once $\gamma > \frac{1}{2}$; Borel–Cantelli concludes. □

Lemma 2. *For N large, $c_{N,2} \gtrsim \log N$.*

Sketch. Expanding (1) and integrating the heat kernels in time,

$$c_{N,2} \gtrsim \sum_{|\omega_i| \le N} \frac{1}{\langle \omega_1 \rangle^2 \langle \omega_2 \rangle^2} \cdot \frac{1}{\langle \omega_1 \rangle^2 + \langle \omega_2 \rangle^2 + \langle \omega_1 + \omega_2 \rangle^2} \gtrsim \sum_{|\omega_1| \le |\omega_2| \le N} \frac{1}{(1 + |\omega_1|^2)(1 + |\omega_2|^4)}.$$

Bounding the sum below by an integral gives $\int_0^N \frac{r}{1+r^2}\, dr \simeq \log N$. □

The following standard facts are recorded for reference.

Lemma 3. (i) *For any polynomial P, $\mathbf{l}_N P(\mathbf{l}_N)$ converges a.s. in $\mathcal{C}^{-1/2-\kappa}$.* (ii) *The expressions $:\mathbf{l}_N^2: \mathbf{C}_N - 3c_{N,2}\mathbf{C}_N$ and $:\mathbf{l}_N^2: v_N + 3c_{N,2}(v_N - \mathbf{C}_N)$ both converge a.s. in $\mathcal{C}^{-1-2\kappa}$ as $N \to \infty$.*

Step 1: $\mu(B_\gamma) = 1$

Fix $\gamma > \frac{1}{2}$. Let u be as in (2) and write $u_N = P_N u$. Expanding the Wick cube and using $u_N = \mathbf{l}_N - \mathbf{C}_N + v_N$,

$$\begin{aligned} (\log N)^{-\gamma} H_3(u_N; c_N) &= (\log N)^{-\gamma} :\mathbf{l}_N^3: - 3(\log N)^{-\gamma} :\mathbf{l}_N^2: \mathbf{C}_N \\ &\quad + 3(\log N)^{-\gamma} :\mathbf{l}_N^2: v_N + (\log N)^{-\gamma} R_N, \end{aligned}$$

where R_N collects the remaining mixed terms. By Lemma 1 the first term $\to 0$; by Lemma 3(i) and standard product estimates the terms in $R_N \to 0$. It remains to treat $-:\mathbf{l}_N^2: \mathbf{C}_N + :\mathbf{l}_N^2: v_N + 3c_{N,2} u_N$, which we regroup as

$$\Big(:\mathbf{l}_N^2: v_N + 3c_{N,2}(v_N - \mathbf{C}_N) \Big) - \Big(:\mathbf{l}_N^2: \mathbf{C}_N - 3c_{N,2}\mathbf{C}_N \Big).$$

By Lemma 3(ii) this converges a.s. in $\mathcal{C}^{-1-2\kappa}$; multiplied by $(\log N)^{-\gamma} \to 0$ it tends to 0. Pairing with ψ,

$$\big\langle (\log N)^{-\gamma} \big(H_3(u_N; c_N) + 9 c_{N,2} u_N \big), \psi \big\rangle \xrightarrow[N\to\infty]{} 0 \quad \text{a.s.},$$

i.e. $u(t) \in B_\gamma$ for fixed t. Hence $\mu(B_\gamma) = 1$.

Step 2: $(T_\psi^*\mu)(B_\gamma) = 0$

We must show $u+\psi \notin B_\gamma$. Expanding with $\psi_N = P_N\psi$,

$$
\begin{aligned}
&(\log N)^{-\gamma} H_3(u_N+\psi_N; c_N) + (\log N)^{-\gamma} 9c_{N,2}(u_N+\psi_N) \\
&\quad = \Big[(\log N)^{-\gamma} H_3(u_N; c_N) + 9(\log N)^{-\gamma} c_{N,2} u_N\Big] \\
&\qquad + 3(\log N)^{-\gamma} {:}u_N^2{:}\, \psi_N + 3(\log N)^{-\gamma} u_N \psi_N^2 \\
&\qquad + (\log N)^{-\gamma}\psi_N^3 + 9(\log N)^{-\gamma} c_{N,2}\psi_N .
\end{aligned}
$$

The bracketed term tends to 0 a.s. (Step 1). Since ${:}u_N^2{:}$ and u_N converge to finite distributional limits (Lemma 3(i)), the next three terms tend to 0 after multiplication by $(\log N)^{-\gamma}$. The decisive term is the last: by Lemma 2,

$$
9(\log N)^{-\gamma} c_{N,2}\, \langle \psi_N, \psi\rangle \;\gtrsim\; (\log N)^{-\gamma+1}\, \langle \psi_N, \psi\rangle .
$$

As $N\to\infty$, $\langle \psi_N,\psi\rangle \to \|\psi\|_{L^2}^2 > 0$ (because $\psi \not\equiv 0$), and $(\log N)^{-\gamma+1}\to\infty$ for $\gamma<1$. Choosing $\frac{1}{2}<\gamma<1$, the displayed quantity *diverges*, so the full pairing does not converge to 0; that is, $u+\psi\notin B_\gamma$ a.s. Therefore $(T_\psi^*\mu)(B_\gamma) = \mu(\{u : u+\psi\in B_\gamma\}) = 0$.

Conclusion

By Steps 1 and 2, $\mu(B_\gamma)=1$ while $(T_\psi^*\mu)(B_\gamma)=0$. Thus μ and $T_\psi^*\mu$ are concentrated on disjoint Borel sets and are mutually singular. The only structural inputs were that $\psi\not\equiv 0$ (used solely to ensure $\|\psi\|_{L^2}^2>0$) and that the coupling is nonzero, so the conclusion holds for all such ψ. $\square$

Remarks

1. **Where dimension three enters.** The entire effect is carried by $c_{N,2} = \mathbb{E}[\mathbf{q}_N \mathbf{C}_N] \gtrsim \log N$ (Lemma 2). This logarithmic divergence is absent in $d=2$, where μ *is* quasi-invariant under smooth shifts and even equivalent to the free field. The borderline dimension for shift-quasi-invariance is exactly 3, and the question is on which side it falls; the answer is the singular side.

2. **Why the heuristic "density" argument fails.** Formally $dT_\psi^*\mu/d\mu$ involves a term proportional to the would-be field $\mathbf{l}^3 - C\mathbf{l}$, which does *not* define a random distribution in $d=3$ (its covariance behaves like $|x-y|^{-3}$, non-integrable). The log-divergent constant in the formal density is precisely what makes the candidate density blow up. The rigorous proof above replaces the nonexistent density by the cut-off functional defining B_γ and extracts the divergence cleanly.

3. **Role of γ.** The window $\frac{1}{2}<\gamma<1$ is forced: $\gamma>\frac{1}{2}$ kills the cube and the chaos remainders (Lemma 1), while $\gamma<1$ lets the shift term $(\log N)^{-\gamma}c_{N,2} \asymp (\log N)^{1-\gamma}$ diverge. Any γ in this range distinguishes the measures.

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Problem 2

Solution to Problem 2: A Universal Test Vector
for Rankin–Selberg Integrals

Statement and Answer

Let F be a non-archimedean local field with ring of integers $\mathfrak{o}$, maximal ideal $\mathfrak{p}$, residue cardinality q, and a nontrivial additive character $\psi : F \to \mathbb{C}^\times$ of conductor $\mathfrak{o}$. Write $G_r := \mathrm{GL}_r(F)$, let $N_r < G_r$ be the upper-triangular unipotent subgroup, and embed $G_n \hookrightarrow G_{n+1}$ as the upper-left block. Set $K_r := \mathrm{GL}_r(\mathfrak{o})$, with Haar measures giving K_r and $N_r \cap K_r$ volume one.

Theorem. *Let Π be a generic irreducible admissible representation of G_{n+1}, realized in its ψ^{-1}-Whittaker model $\mathcal{W}(\Pi, \psi^{-1})$. Then there exists a single $W \in \mathcal{W}(\Pi, \psi^{-1})$ with the following property. For every generic irreducible admissible π of G_n with conductor ideal $\mathfrak{q}$, generator Q of $\mathfrak{q}^{-1}$, and $u_Q := I_{n+1} + QE_{n,n+1}$, there exists $V \in \mathcal{W}(\pi, \psi)$ such that*

$$\int_{N_n \backslash G_n} W(\mathrm{diag}(g, 1)u_Q)\, V(g)\, |\det g|^{s-\frac{1}{2}}\, dg$$

is finite and nonzero for all $s \in \mathbb{C}$.

The answer is **YES**. The single W may be taken to be an explicit, Π-independent-of-π Kirillov-model vector, and for each π the role of V is played by the *normalized newvector*. The integral is in fact *constant in s* and equal to an explicit nonzero scalar; this is why finiteness and nonvanishing hold for all s, with no analytic continuation or pole-cancellation required.

A failure mode to avoid. One cannot choose V as a Schwartz bump in the Kirillov model making the integrand constant on its support: V has a central character, so it cannot be compactly supported modulo the center. (Already for $n = 1$ the relevant integral is a normalized Gauss sum, whose integrand is genuinely non-constant.) The correct argument runs through the Godement–Jacquet functional equation, not through test-vector flexibility.

Choice of W and V

For $x \in F$ and $y \in F^\times$ write

$$d_y := \mathrm{diag}(1, \dots, 1, y) \in G_n \hookrightarrow G_{n+1}, \qquad u_x := I_{n+1} + xE_{n,n+1} \in N_{n+1}.$$

The vector W. Define W_0 on G_n by

$$W_0(g) := \int_{N_n} \mathbf{1}_{K_n}(xg)\, \psi(x)\, dx, \tag{4}$$

a ψ^{-1}-Whittaker function on G_n which extends, via the theory of the Kirillov model [1], to an element of $\mathcal{W}(\Pi, \psi^{-1})$ on G_{n+1}. We take $W := u_Q W_0$ (equivalently we evaluate W_0 at the shifted argument); W_0 depends only on Π, not on π.

The vector V. Given π with conductor $\mathfrak{q}$, let $V \in \mathcal{W}(\pi, \psi)$ be the normalized newvector: the unique $K_1(\mathfrak{q})$-invariant vector with $V(1) = 1$, where $K_1(\mathfrak{q}) = \{g \in K_n : \text{last row} \equiv (0, \dots, 0, 1) \bmod \mathfrak{q}\}$ [4, 5].

We choose a generator Q of $\mathfrak{q}^{-1}$, so $|Q| = [\mathfrak{o} : \mathfrak{q}]$. Recall the conductor is characterized by the ε-factor:

$$\varepsilon(\tfrac{1}{2} + s, \pi, \psi) = |Q|^{-s}\, \varepsilon(\tfrac{1}{2}, \pi, \psi). \tag{5}$$

The Godement–Jacquet engine

Lemma 1 (Godement–Jacquet functional equation [3])**.** *Let f be a matrix coefficient of π and $\phi \in \mathcal{S}(M_n(F))$. The zeta integral*

$$Z(\phi, f, s) := \int_{G_n} \phi(g) f(g) \, |\det g|^{\frac{n-1}{2}+s} \, dg$$

converges for $\Re(s)$ large, continues meromorphically with $Z(\phi, f, s)/L(s, \pi)$ holomorphic, and satisfies

$$\gamma(s, \pi, \psi) \, Z(\phi, f, s) = Z(\widehat{\phi}, f^\vee, 1-s), \qquad \gamma(s, \pi, \psi) = \varepsilon(s, \pi, \psi) \frac{L(1-s, \widetilde{\pi})}{L(s, \pi)},$$

where $f^\vee(g) = f(g^{-1})$ and $\widehat{\phi}$ is the ψ-self-dual Fourier transform on $M_n(F)$. The identities persist when f is taken to be a Whittaker function of π.

Let $\mathcal{S}_{\mathrm{e}}(F^\times)$ be the Schwartz–Bruhat functions β with $\beta(xy) = \beta(x)$ whenever $|y| = 1$ (i.e. β depends only on $|x|$), with Mellin inversion $\beta(y) = \int_{(\sigma)} \tilde{\beta}(s)|y|^s \, ds$. For such β define $\beta^\sharp$ by $\beta^\sharp(y) = \int_{(\sigma)} \tilde{\beta}(s)|y|^{-s} \, ds/\gamma(\frac{1}{2} + s, \pi, \psi)$.

Lemma 2. *Define β by $\tilde{\beta}(s) = \varepsilon(\frac{1}{2} + s, \pi, \psi)/L(\frac{1}{2} + s, \pi)$. Then:*

1. *β is supported on $\{|Q| \le |y| \le |Q|q^n\}$ and equals $\varepsilon(\frac{1}{2}, \pi, \psi)$ on $\{|y| = |Q|\}$;*
2. *$\beta^\sharp$ has Mellin transform $\widetilde{\beta^\sharp}(s) = 1/L(\frac{1}{2} + s, \widetilde{\pi})$, is supported on $\{1 \le |y| \le q^n\}$, and equals 1 on $\mathfrak{o}^\times = \{|y| = 1\}$.*

Proof. By definition of γ, $\widetilde{\beta^\sharp}(s) = \tilde{\beta}(s)/\gamma(\frac{1}{2}+s, \pi, \psi) = L(\frac{1}{2}-s, \widetilde{\pi})^{-1} \cdot (\cdots)$ simplifies to $1/L(\frac{1}{2}+s, \widetilde{\pi})$. The inverse L-values are monic polynomials in q^{-s} of degree $\le n$, so by Mellin inversion and (5), β and $\beta^\sharp$ have the stated supports and boundary values. $\square$

Lemma 3. *For π unitary generic, with f a Whittaker function and $\phi \in \mathcal{S}(M_n(F))$,*

$$\int_{G_n} \phi(g) f(g) \beta(\det g) |\det g|^{n/2} \, dg = \int_{G_n} \widehat{\phi}(g) f^\vee(g) \beta^\sharp(\det g) |\det g|^{n/2} \, dg.$$

Proof. Insert the Mellin expansion of β at $\sigma = 0$; the resulting double integral converges absolutely, and is recognized as $\int_{(0)} \tilde{\beta}(s) Z(\phi, f, \frac{1}{2} + s) \, ds$. Apply the functional equation (Lemma 1) and recognize the result as the right-hand side. $\square$

The local computation

Set $t_Q := d_Q^{-1} u_Q = u_1 d_Q^{-1}$.

Lemma 4. *There exist $\beta \in \mathcal{S}_{\mathrm{e}}(F^\times)$ and $\phi \in \mathcal{S}(M_n(F))$ with, for all $g \in G_n$,*

$$\int_{N_n} \beta(\det xg) \, \phi(xg) \, \psi(x) \, dx = \varepsilon(\tfrac{1}{2}, \pi, \psi) \, W_0(g \, t_Q), \tag{6}$$

$$\beta^\sharp(\det g) \, \widehat{\phi}(g) = |Q|^n \, \mathbf{1}_{K_1(\mathfrak{q})}(g). \tag{7}$$

Proof. Take β as in Lemma 2 and $\phi(x) := \psi(-x_{nn})\phi_0(xd_Q^{-1})$ with $\phi_0 = \mathbf{1}_{M_n(\mathfrak{o})}$. For (6), use $W_0(gt_Q) = \psi(-g_{nn})W_0(gd_Q^{-1})$ and the identities $\phi(xg) = \psi(-g_{nn})\phi_0(xgd_Q^{-1})$, $\beta(\det g)\phi_0(gd_Q^{-1}) = \varepsilon(\frac{1}{2},\pi,\psi)\mathbf{1}_{K_n}(gd_Q^{-1})$ (the latter from Lemma 2(1) and $\mathbf{1}_{K_n}(g) = \mathbf{1}_{\mathfrak{o}^\times}(\det g)\phi_0(g)$), then integrate against $\psi(x)\,dx$ recalling (4). For (7), the Fourier computation $\widehat{\phi}(x) = |Q|^n\widehat{\phi_0}(d_Q(x - E_{nn}))$ with $\widehat{\phi_0} = \phi_0$, together with Lemma 2(2), gives $\beta^\sharp(\det g)\widehat{\phi}(g) = |Q|^n\mathbf{1}_{K_1(\mathfrak{q})}(g)$, since $K_1(\mathfrak{q}) = \{g : d_Q(g - E_{nn}) \in M_n(\mathfrak{o}),\ \det g \in \mathfrak{o}^\times\}$. $\square$

Conclusion

For $W \in \mathcal{W}(\Pi, \psi^{-1})$, $V \in \mathcal{W}(\pi, \psi)$, write the Rankin–Selberg integral

$$\ell_{\mathrm{RS}}(s, W, V) := \int_{N_n\backslash G_n} W(\mathrm{diag}(g,1))\,V(g)\,|\det g|^{s-\frac{1}{2}}\,dg.$$

Proposition 1. *Let W_0 be as in (4) and V the normalized newvector of π. Then for all $s \in \mathbb{C}$,*

$$\ell_{\mathrm{RS}}(s, u_QW_0,\, d_QV) = c\,|Q|^{-n/2}, \qquad c := \varepsilon(\tfrac{1}{2},\pi,\psi)^{-1}|Q|^n\,\mathrm{vol}(K_1(\mathfrak{q})) \asymp 1.$$

Proof. A change of variables gives the homogeneity $\ell_{\mathrm{RS}}(s, u_QW_0, d_QV) = |Q|^{-(s-\frac{1}{2})}\ell_{\mathrm{RS}}(s, t_QW_0, V)$. Since W_0 is supported on $\det^{-1}(\mathfrak{o}^\times)$, the translate t_QW_0 is supported on $\det^{-1}(Q\mathfrak{o}^\times)$, so $\ell_{\mathrm{RS}}(s, t_QW_0, V)$ is a constant multiple of $|Q|^s$; it therefore suffices to evaluate at $s = \frac{n+1}{2}$. Unfolding and applying Lemmas 4 (i.e. (6)), 3, and (7) in turn, with $f(g) = V(g)$,

$$\begin{aligned}
\varepsilon(\tfrac{1}{2},\pi,\psi)\,\ell_{\mathrm{RS}}(\tfrac{n+1}{2}, t_QW_0, V) &= \varepsilon(\tfrac{1}{2},\pi,\psi)\int_{N_n\backslash G_n} W_0(gt_Q)V(g)|\det g|^{n/2}dg \\
&= \int_{G_n} \phi(g)f(g)\beta(\det g)|\det g|^{n/2}dg \\
&= \int_{G_n} \widehat{\phi}(g)f^\vee(g)\beta^\sharp(\det g)|\det g|^{n/2}dg \\
&= |Q|^n\int_{K_1(\mathfrak{q})} V(g^{-1})|\det g|^{n/2}dg = |Q|^n\,\mathrm{vol}(K_1(\mathfrak{q})),
\end{aligned}$$

using $K_1(\mathfrak{q})$-invariance, $V(1) = 1$, and $|\det g| = 1$ on $K_1(\mathfrak{q})$. Hence $\ell_{\mathrm{RS}}(\frac{n+1}{2}, t_QW_0, V) = c$, so $\ell_{\mathrm{RS}}(s, t_QW_0, V) = c\,|Q|^{s-\frac{n+1}{2}}$ and $\ell_{\mathrm{RS}}(s, u_QW_0, d_QV) = c\,|Q|^{-n/2}$. $\square$

Since $c\,|Q|^{-n/2}$ is a nonzero scalar independent of s, the integral is finite and nonzero for all $s \in \mathbb{C}$. The vector $W := u_QW_0$ (built from the single, π-independent W_0) and the π-newvector V (rescaled by d_Q) exhibit the required test vectors. This proves the Theorem. $\square$

Remarks

The mechanism is transparent: u_QW_0 is supported on a single det-coset, so the zeta integral collapses to a *constant* in s rather than an L-factor times a unit. The Godement–Jacquet functional equation, packaged through the γ-factor transform $\beta \mapsto \beta^\sharp$ (Lemma 2), converts the ψ-twisted unipotent shift into an integral over a translate of $K_1(\mathfrak{q})$, where newvector theory yields $\mathrm{vol}(K_1(\mathfrak{q}))$. The $n = 1$ case recovers the classical normalized Gauss sum and has appeared in subconvexity and moment applications [7, 6]; the general n statement, with a single unipotent translate rather than an average, is the present lemma (joint work in progress with S. Jana). A version with an average over many translates appears in [2].

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Problem 3

Solution to Problem 3: A Probabilistic Interpretation
for Interpolation Macdonald Polynomials

Statement and Answer

Let $\lambda = (\lambda_1 > \cdots > \lambda_n \geq 0)$ be a partition with distinct parts that is moreover *restricted*: it has a unique part of size 0 and no part of size 1. Let $S_n(\lambda)$ be the orbit of λ under permutation of parts (equivalently, the compositions that rearrange λ). Let $F^*_\mu(x_1, \dots, x_n; q, t)$ and $P^*_\lambda(x_1, \dots, x_n; q, t)$ be the interpolation ASEP polynomial and interpolation Macdonald polynomial.

Theorem. *There is a nontrivial Markov chain on $S_n(\lambda)$ — the* interpolation t-Push TASEP *— whose stationary distribution is*

$$\pi^*_\lambda(\mu) \; = \; \frac{F^*_\mu(x_1, \dots, x_n;\, q = 1,\, t)}{P^*_\lambda(x_1, \dots, x_n;\, q = 1,\, t)}, \qquad \mu \in S_n(\lambda),$$

and whose transition probabilities are not described using the polynomials F^*_μ.

The answer is **YES**. We caution at the outset that the chain is *not* the ordinary multispecies ASEP: that chain has the ordinary (not interpolation) Macdonald polynomials as its stationary weights (Ayyer–Martin–Williams), and substituting it here would solve a different, already-known problem. The correct object is a $q = 1$ *push*-TASEP with an interpolation correction encoded by a ringing "bell".

Probabilities and notation

Throughout we work at $q = 1$ and write $P^*_\lambda = P^*_\lambda(x; 1, t)$, etc. For $1 \leq k \leq n$ put

$$p_k := \frac{t^{-n+1}(1-t)}{x_k - t^{-n+2}}, \qquad q_k := \frac{(1-t)\, x_k}{x_k - t^{-n+2}}.$$

For $0 < t < 1$ and $x_i > t^{-n+1}$ these lie in $[0, 1]$ and serve as skipping/displacement probabilities. We use $e^*_k(x; t)$ for the interpolation elementary symmetric polynomials and the factorization (valid at $q = 1$ [6, 3])

$$P^*_\lambda(x; 1, t) = \prod_{1 \leq i \leq \lambda_1} e^*_{\lambda'_i}(x; t), \tag{8}$$

λ' the conjugate partition.

The interpolation t-Push TASEP

Definition 1. Fix λ with $\lambda_n = 0$. States of the chain are configurations of particles on a ring of n sites labelled by the parts of λ: state η places the particle labelled η_j at site j (label 0 is a vacancy). One step from η proceeds in three stages.

(Step 0) The bell. The bell at site j rings with probability

$$P_j = \frac{\prod_{k<j}(x_k - t^{-n+2}) \prod_{k>j}(x_k - t^{-n+1})}{e^*_{n-1}(x; t)}, \quad e^*_{n-1}(x; t) = \sum_{j=1}^{n} \prod_{k<j}(x_k - t^{-n+2}) \prod_{k>j}(x_k - t^{-n+1}).$$

(Step 1) Push-TASEP move. The particle at site j, say with label a, is activated and travels clockwise by the t-Push TASEP rule: if there are m weaker particles in the system (labels $< a$, vacancies counting as label 0), the active particle moves to the k-th such weaker particle with probability $t^{k-1}/[m]_t$. If that site held a positive label, that particle becomes active and repeats; the cascade ends when the active particle reaches a vacancy. Afterwards site j is vacant; we treat this vacancy as a new particle labelled $a := 0$.

(Step 2) Vacancy sweep. The label-0 particle goes to site 1 and travels clockwise. Reaching a site k ($1 \le k \le j-1$) holding a label $b \ge 0$, it skips that site with probability $1 - p_k$ if $b \ge a$ and $1 - q_k$ if $b < a$; otherwise it settles there, activating/displacing the resident, which then continues clockwise toward site j by the same rule. The active particle stops once it displaces another or arrives at site j, where it settles.

We denote the transition probabilities of Steps 1 and 2 by $P_j^{(1)}$ and $P_j^{(2)}$, so that for $\mu, \nu \in S_n(\lambda)$,

$$P(\mu,\nu) = \sum_{1\le j\le n} P_j \sum_{\rho\in S_n(\lambda):\rho_j=0} P_j^{(1)}(\mu,\rho)\, P_j^{(2)}(\rho,\nu).$$

Remark 1 (nontriviality)**.** The rates P_j, p_k, q_k are explicit rational functions of $x_1,\dots,x_n$ and t only. They do *not* reference F^*_μ, so the chain is nontrivial in the required sense (unlike a Metropolis–Hastings chain built from the target distribution).

Theorem 1. *In the interpolation t-Push TASEP with content λ, the stationary probability of $\mu \in S_n(\lambda)$ is $\pi^*_\lambda(\mu) = F^*_\mu(x;1,t)/P^*_\lambda(x;1,t)$.*

Two-line queues and the proof

We use classical two-line queues [5] and signed two-line queues [3], specialized to $q = 1$. Let $\mathcal{Q}^\eta_\kappa$ be the classical two-line queues with top row η and bottom row κ, with pair-weight generating function $a^\eta_\kappa = \sum_{Q\in\mathcal{Q}^\eta_\kappa} \mathrm{wt}_{\mathrm{pair}}(Q)$. Let $\mathcal{G}^\alpha_\mu$ be the signed two-line queues with top row $\alpha \in \mathbb{Z}^n$ and bottom row μ, with $b^\alpha_\mu = \sum_{Q\in\mathcal{G}^\alpha_\mu} \mathrm{wt}_{\mathrm{pair}}(Q)$, and full weight $\mathrm{wt}(Q) = \mathrm{wt}_{\mathrm{pair}}(Q)\,\mathrm{wt}_{\mathrm{ball}}(Q)$ where

$$\mathrm{wt}_\alpha\, b^\alpha_\mu = \sum_{Q\in\mathcal{G}^\alpha_\mu} \mathrm{wt}(Q), \quad \mathrm{wt}_\alpha := \prod_{k:\alpha_k>0} x_k \prod_{k:\alpha_k<0} \left(-t^{-(n-1)}\right). \tag{9}$$

Definition 2. For a signed two-line queue $Q \in \mathcal{G}^\alpha_\mu$ let $\overline{Q}$ be its unsigned version (forget signs on the top row); its top composition is $\|\alpha\| = (|\alpha_1|,\dots,|\alpha_n|)$. Let $\mathcal{G}^\kappa_\mu$ be the set of unsigned objects so obtained, with $\|\alpha\| = \kappa$. For $\overline{Q} \in \mathcal{G}^\kappa_\mu$ assign a nontrivial pairing p the weight $(1-t)t^{\mathrm{skip}(p)}$, and a top-row ball B in column k labelled $a > 0$ with the ball below labelled b the weight

$$\mathrm{wt}(B) = \begin{cases} x_k - t^{-(n-1)} & b = a, \\ x_k & b > a, \\ t^{-(n-1)} & b < a. \end{cases}$$

Lemma 1. *For λ with distinct parts and $\kappa,\mu \in S_n(\lambda)$, and $\overline{Q} \in \mathcal{G}^\kappa_\mu$, $\mathrm{wt}(\overline{Q}) = \sum_Q \mathrm{wt}(Q)$, the sum over signed Q forgetting to $\overline{Q}$.*

Proof. Enumerate the admissible sign choices for each top-row ball B labelled $a > 0$: a $-$ sign is forced when the ball below is a vacancy or labelled $b < a$; a $+$ sign is forced when $b > a$; both are allowed when $b = a$. One checks this matches (9): a $-$ sign multiplies the ball weight by -1 in passing from Q to $\overline{Q}$, but the weight of the pairing attached to B is also multiplied by -1, so signs cancel consistently. $\square$

For $\kappa \in S_n(\lambda)$ set $c^\kappa_\mu := \sum_{\alpha:\|\alpha\|=\kappa} \mathrm{wt}_\alpha\, b^\alpha_\mu$. Combining (9) and Lemma 1:

Lemma 2. $c^\kappa_\mu = \sum_{\overline{Q}\in\mathcal{G}^\kappa_\mu} \mathrm{wt}(\overline{Q})$. *Since λ has distinct parts, $\mathcal{G}^\kappa_\mu$ is empty or a single element.*

The closing recursion (where the hypotheses on λ enter)

Fix a weakly order-preserving $\phi : \mathbb{N} \to \mathbb{N}$ and partitions λ, κ with $\phi(\lambda) = \kappa$. For $\eta \in S_n(\kappa)$ set $G^*_\eta(x;t) := \sum_{\rho\in S_n(\lambda):\,\phi(\rho)=\eta} F^*_\rho(x;1,t)$. The following is an interpolation analogue of a result of Ayyer–Martin–Williams [2], provable in the same way using [1].

Theorem 2. *For λ, κ as above and all $\eta \in S_n(\kappa)$, at $q = 1$,*

$$\frac{G^*_\eta(x;t)}{P^*_\lambda(x;1,t)} = \frac{F^*_\eta(x;1,t)}{P^*_\kappa(x;1,t)}.$$

Corollary 1. *Let ρ be a composition with $\rho_i \neq 1$ for all i, with k nonzero parts, and set $\eta = \rho^-$ (where $\rho^-_i = \max(\rho_i - 1, 0)$). Then at $q = 1$,*

$$F^*_\rho(x;1,t) = F^*_\eta(x;1,t) \cdot e^*_k(x;t).$$

Proof. Let λ, κ reorder ρ, η. Take $\phi : i \mapsto \max(i-1, 0)$, so $\phi(\rho) = \eta$. *Because λ has no part of size* 1, ϕ is a bijection from $\{0, 2, 3, \dots\}$ onto $\{0, 1, 2, \dots\}$, so ρ is the unique composition in $S_n(\lambda)$ with $\phi(\rho) = \eta$; hence $G^*_\eta = F^*_\rho$. By Theorem 2, $F^*_\rho/P^*_\lambda = F^*_\eta/P^*_\kappa$. Finally, by (8) and the fact that κ is obtained from λ by deleting the largest column (of size k), $P^*_\lambda/P^*_\kappa = e^*_k(x;t)$, giving the claim. □

This is exactly where the restriction on λ is essential: *no part of size* 1 makes ϕ bijective so the recursion closes, and the *unique part of size* 0 guarantees η has a unique 0-part so the column-deletion is unambiguous. The hypotheses are structural, not (as one might guess) a device to avoid a 0/0 normalization.

From two-line queues to the dynamics

Proposition 1. *For $\rho, \nu \in S_n(\lambda)$ with $\rho_j = 0$,*

$$P^{(2)}_j(\rho,\nu) = \frac{c^\rho_\nu}{\prod_{k<j}(x_k - t^{-n+2})\prod_{k>j}(x_k - t^{-n+1})}, \quad \textit{equivalently} \quad P_j\, P^{(2)}_j(\rho,\nu) = \frac{c^\rho_\nu}{e^*_{n-1}}.$$

Idea. Step 2 is encoded by the unique element of $\mathcal{G}^\rho_\nu$ (Lemma 2): a particle labelled a moving from column k to k' corresponds to a nontrivial pairing joining a top ball in column k to a bottom ball in column k'; stationary particles give trivial pairings. Dividing each ball/pairing weight in $\mathrm{wt}(\overline{Q})$ by the matching factor of $D = \prod_{k<j}(x_k - t^{-n+2})\prod_{k>j}(x_k - t^{-n+1})$ reproduces, case by case, the skip/settle probabilities $1-p_k$, $1-q_k$, p_k, q_k of Step 2 (and a factor 1 for $k > j$, which never moves). Hence $\mathrm{wt}(\overline{Q})/D = P^{(2)}_j(\rho,\nu)$. □

Proposition 2. *If λ is restricted and $\mu, \nu \in S_n(\lambda)$, then* $P(\mu,\nu) = \displaystyle\sum_{\rho\in S_n(\lambda)} \frac{a^\mu_\rho\, c^\rho_\nu}{e^*_{n-1}}$.

Proof. Combine the encoding of Step 1 by classical two-line queues [2, Lem. 5.4] (giving $P^{(1)}_j(\mu,\rho)$ via a^μ_ρ) with Proposition 1:

$$P(\mu,\nu) = \sum_j P_j \sum_{\rho:\rho_j=0} P^{(1)}_j(\mu,\rho) P^{(2)}_j(\rho,\nu) = \sum_j \sum_{\rho:\rho_j=0} \frac{a^\mu_\rho c^\rho_\nu}{e^*_{n-1}} = \sum_{\rho\in S_n(\lambda)} \frac{a^\mu_\rho c^\rho_\nu}{e^*_{n-1}}.$$

□

Proof of Theorem 1

Proof. Fix $\nu \in S_n(\lambda)$. By [3, Thm. 1.15, Lem. 5.6],

$$F^*_\nu(x;1,t) = \sum_{\eta\in\mathbb{N}^n} F^{*\eta}_\nu(x;t)\, F^*_{\eta^-}(x;1,t), \qquad F^{*\eta}_\nu(x;t) = \sum_{\kappa\in\mathbb{N}^n} a^\eta_\kappa\, c^\kappa_\nu.$$

By Corollary 1 (using that η has a unique 0-part), $F^*_{\eta^-}(x;1,t) = F^*_\eta(x;1,t)/e^*_{n-1}(x;t)$. Substituting,

$$F^*_\nu(x;1,t) = \sum_{\eta\in\mathbb{N}^n} F^*_\eta(x;1,t) \sum_{\kappa\in\mathbb{N}^n} \frac{a^\eta_\kappa\, c^\kappa_\nu}{e^*_{n-1}(x;t)} = \sum_{\eta\in\mathbb{N}^n} F^*_\eta(x;1,t)\, P(\eta,\nu),$$

the last equality by Proposition 2. Thus the vector $\big(F^*_\mu(x;1,t)\big)_\mu$ is left-fixed by the transition matrix, i.e. proportional to the stationary distribution. Since $P^*_\lambda = \sum_{\mu\in S_n(\lambda)} F^*_\mu$, normalizing gives $\pi^*_\lambda(\mu) = F^*_\mu(x;1,t)/P^*_\lambda(x;1,t)$. □

Remark on the failure mode to avoid

Replacing this chain by the ordinary multispecies ASEP (matrix-product ansatz / Zamolodchikov–Faddeev algebra) does *not* solve the stated problem: that construction yields the ordinary Macdonald polynomials as stationary weights [5, 2], not the *interpolation* polynomials F^*_μ, P^*_λ. The interpolation correction is precisely what the bell P_j and the push-with-sweep dynamics supply, and the proof above runs through signed two-line queues rather than a trace over an auxiliary algebra.

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Problem 4

Solution to Problem 4: The Finite Free Stam Inequality

Statement and Answer

For a monic real-rooted polynomial p of degree n with roots $\alpha = (\alpha_1, \dots, \alpha_n)$, define the *finite free score vector* $\mathsf{S}_n^{\mathrm{ff}}(\alpha) \in (\mathbb{R} \cup \{\infty\})^n$ by

$$\mathsf{S}_n^{\mathrm{ff}}(\alpha)_i \;=\; \sum_{j \neq i} \frac{1}{\alpha_i - \alpha_j},$$

and the *finite free Fisher information* $\mathcal{K}_n^{\mathrm{ff}}(p) := \|\mathsf{S}_n^{\mathrm{ff}}(\alpha)\|^2$, with $\mathcal{K}_n^{\mathrm{ff}}(p) = \infty$ if p has a multiple root. Let $\boxplus_n$ denote the finite free convolution of degree-n polynomials.

Theorem. *For any two monic real-rooted polynomials p, q of degree n,*

$$\frac{1}{\mathcal{K}_n^{\mathrm{ff}}(p \boxplus_n q)} \;\geq\; \frac{1}{\mathcal{K}_n^{\mathrm{ff}}(p)} + \frac{1}{\mathcal{K}_n^{\mathrm{ff}}(q)}.$$

The answer is **YES**. The result was conjectured by D. Shlyakhtenko; the proof below follows Blachman's strategy for the classical Stam inequality, with the key analytic input replaced by a convexity statement for hyperbolic polynomials.

Preliminaries on $\boxplus_n$

We recall the standard description of the finite free convolution. If α and β are vectors of roots for p and q, then

$$p(x) \boxplus_n q(x) \;=\; \sum_{\pi \in S_n} \prod_{i=1}^{n} \big(x - \alpha_i - \beta_{\pi(i)}\big), \tag{10}$$

and Walsh's theorem guarantees that $p \boxplus_n q$ is again real-rooted [2, 1]. Thus $\boxplus_n$ induces a map

$$\Psi_{\boxplus_n} : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n,$$

where $\Psi_{\boxplus_n}(\alpha, \beta)$ is the ordered root vector of $p \boxplus_n q$. Write $m_k(\alpha) = \frac{1}{n}\sum_i \alpha_i^k$ and $\mathrm{Var}(\alpha) = m_2(\alpha) - m_1(\alpha)^2$, and let $\mathbf{1}_n$ be the all-ones vector.

Proposition 1 (properties of $\boxplus_n$)**.** *If $\gamma = \Psi_{\boxplus_n}(\alpha, \beta)$ then*

1. *(additivity) $m_1(\gamma) = m_1(\alpha) + m_1(\beta)$ and $\mathrm{Var}(\gamma) = \mathrm{Var}(\alpha) + \mathrm{Var}(\beta)$;*
2. *(translation) $\Psi_{\boxplus_n}(\alpha + t\mathbf{1}_n, \beta) = \gamma + t\mathbf{1}_n$ and likewise in the second argument.*

Proof. (i) follows from (10) together with the Newton identities applied to the first two coefficients of $p \boxplus_n q$; (ii) is immediate from (10). ☐

Score vectors via the heat flow

The link between $\mathcal{K}_n^{\mathrm{ff}}$ and the dynamics of roots is the following, due to Shlyakhtenko.

Lemma 1 (score vector as a velocity)**.** *Assume p has simple roots. Let $p_t(x) := \exp\bigl(-\frac{t}{2}\partial_x^2\bigr)p(x)$ and let $\alpha(t)$ be the ordered root vector of p_t. Then $\alpha_i'(0) = \sum_{j\neq i} \frac{1}{\alpha_i-\alpha_j}$; that is, $\alpha'(0) = \mathsf{S}_n^{\mathrm{ff}}(\alpha)$.*

Proof. Roots stay simple near $t = 0$. Implicitly differentiating $p(\alpha_i(t)) - \frac{t}{2}p''(\alpha_i(t)) + t^2 R(\alpha_i(t), t) = 0$ at $t = 0$ gives $\alpha_i'(0) = \frac{1}{2}\, p''(\alpha_i)/p'(\alpha_i)$, which equals $\sum_{j\neq i}(\alpha_i - \alpha_j)^{-1}$. $\square$

The essential point is that $\exp(-\frac{t}{2}\partial_x^2)$ *commutes with* $\boxplus_n$ (finite free convolution commutes with any constant-coefficient differential operator [1]). We exploit this twice.

Reduction to simple roots

Lemma 2. *Let $T_\epsilon := (1-\epsilon\, d/dx)^n$. For monic real-rooted p of degree n: (i) $T_\epsilon p$ is monic real-rooted with simple roots; (ii) $\mathcal{K}_n^{\mathrm{ff}}(T_\epsilon p) \to \mathcal{K}_n^{\mathrm{ff}}(p)$ as $\epsilon \to 0$; (iii) $(T_\epsilon p) \boxplus_n (T_\epsilon q) = T_\epsilon^2(p \boxplus_n q)$.*

Proof. (i) is Nuij's theorem [3]; (ii) holds since $\mathcal{K}_n^{\mathrm{ff}}$ is continuous in the roots and the roots are continuous in ϵ; (iii) holds because $\boxplus_n$ commutes with the differential operator $1 - \epsilon\, d/dx$ [1]. $\square$

By Lemma 2(iii) and (ii), it suffices to prove the Theorem when p, q, and $p \boxplus_n q$ all have simple roots; we assume this henceforth. Then $\alpha, \beta, \gamma = \Psi_{\boxplus_n}(\alpha, \beta)$ have distinct entries and are smooth functions of the coefficients of the corresponding polynomials. Let $\mathsf{D}_{\boxplus_n}$ denote the Jacobian of $\Psi_{\boxplus_n}$ at (α, β).

Step 1: score vectors and the Jacobian

Observation 1. For all $a, b \geq 0$,

$$\mathsf{D}_{\boxplus_n}\bigl(a\,\mathsf{S}_n^{\mathrm{ff}}(\alpha),\, b\,\mathsf{S}_n^{\mathrm{ff}}(\beta)\bigr) \;=\; (a+b)\,\mathsf{S}_n^{\mathrm{ff}}(\gamma).$$

Proof. Set $p_t = \exp(-\frac{t}{2}\partial_x^2)p$ with ordered roots $\alpha(t)$, and define $q_t, \beta(t)$ and $r_t, \gamma(t)$ analogously. Because $\boxplus_n$ commutes with the heat operator,

$$r_{(a+b)t} \;=\; p_{at} \boxplus_n q_{bt},$$

so $\gamma((a+b)t) = \Psi_{\boxplus_n}(\alpha(at), \beta(bt))$. Differentiating at $t = 0$ and using the chain rule,

$$(a+b)\,\gamma'(0) \;=\; \mathsf{D}_{\boxplus_n}\begin{pmatrix} a\,\alpha'(0) \\ b\,\beta'(0) \end{pmatrix}.$$

Lemma 1 identifies $\alpha'(0) = \mathsf{S}_n^{\mathrm{ff}}(\alpha)$, etc., giving the claim. $\square$

Step 2: the Jacobian is a contraction on mean-zero vectors

This is the substance of the proof.

Proposition 2. *If $u, v \in \mathbb{R}^n$ are each orthogonal to $\mathbf{1}_n$, then*

$$\|\mathsf{D}_{\boxplus_n}(u, v)\|^2 \;\leq\; \|u\|^2 + \|v\|^2.$$

Write $\gamma_i = (\Psi_{\boxplus_n})_i(\alpha, \beta)$ for the coordinate functions and $H^{(i)} := \mathrm{Hess}(\Psi_{\boxplus_n})_i$. Let $V = \{(u, v) \in \mathbb{R}^n \times \mathbb{R}^n : u^*\mathbf{1}_n = v^*\mathbf{1}_n = 0\}$.

Lemma 3 (Hessian identity)**.** *For all $w \in V$,*

$$w^*\mathsf{D}_{\boxplus_n}\mathsf{D}_{\boxplus_n}^* w \;=\; w^*\Big(I_n \oplus I_n - \sum_{i=1}^{n} \gamma_i H^{(i)}\Big)w.$$

Proof. Fix $w = (u, v) \in V$ and set $\alpha(t) = \alpha + tu$, $\beta(t) = \beta + tv$, $\gamma(t) = \Psi_{\boxplus_n}(\alpha(t), \beta(t))$. By the additivity of variance (Proposition 1(i)),

$$m_2(\gamma(t)) - m_1(\gamma(t))^2 = m_2(\alpha(t)) + m_2(\beta(t)) - m_1(\alpha(t))^2 - m_1(\beta(t))^2.$$

Since $(u, v) \in V$, the means $m_1(\alpha(t)), m_1(\beta(t))$ — hence $m_1(\gamma(t))$ — are constant in t. Differentiating twice at $t = 0$,

$$\partial_t^2 m_2(\gamma(t))\big|_0 = \partial_t^2\big(m_2(\alpha(t)) + m_2(\beta(t))\big)\big|_0.$$

The left side equals $\frac{1}{n}\sum_i \partial_t^2(\gamma_i^2) = \frac{2}{n}\sum_i\big((\partial_t\gamma_i)^2 + \gamma_i\partial_t^2\gamma_i\big) = \frac{2}{n}\big(w^*\mathsf{D}_{\boxplus_n}\mathsf{D}_{\boxplus_n}^* w + \sum_i \gamma_i w^* H^{(i)} w\big)$, while the right side equals $\frac{2}{n}\, w^*w$. Rearranging gives the identity. □

The minus sign means we must show $\sum_i \gamma_i H^{(i)} \succeq 0$ on V. This is where real-rootedness of $\boxplus_n$ is used in an essential way, via the theory of hyperbolic polynomials.

Theorem 1 (Bauschke–Güler–Lewis–Sendov [5])**.** *Let $f \in \mathbb{R}[x_1, \ldots, x_m]$ be hyperbolic in the direction $w \in \mathbb{R}^m$, and for $a \in \mathbb{R}^m$ let $\lambda_1(a) \geq \cdots \geq \lambda_m(a)$ be the roots of $t \mapsto f(a + tw)$. Then for each k, $\sigma_k(a) := \sum_{i=1}^{k} \lambda_i(a)$ is convex in a.*

Corollary 1. *For any reals $c_1 \geq \cdots \geq c_n$, the matrix $\sum_{i=1}^{n} c_i H^{(i)}$ is PSD.*

Proof. Consider $f(x, a_1, \ldots, a_n, b_1, \ldots, b_n) = \sum_{\pi \in S_n} \prod_{i=1}^{n}(x - a_i - b_{\pi(i)})$. By (10) and Walsh's theorem f is homogeneous and hyperbolic in the direction $e_1 = (1, 0, \ldots, 0)$. By Theorem 1 each $\sigma_k(x, a, b) = \sum_{i=1}^{k} \lambda_i(x, a, b)$ is convex, where $\lambda_i(x, a, b)$ are the roots of $f((x, a, b) + te_1)$. Since the c_i are ordered,

$$L(x, a, b) := \sum_{i=1}^{n} c_i \lambda_i(x, a, b)$$

is a nonnegative combination of the σ_k (by Abel summation), hence convex; so $\mathrm{Hess}\, L = \sum_i c_i \,\mathrm{Hess}\, \lambda_i \succeq 0$ everywhere. Evaluating at $x = 0$, $a = \alpha$, $b = \beta$, we have $\mathrm{Hess}\, \lambda_i = H^{(i)}$, giving $\sum_i c_i H^{(i)} \succeq 0$. □

Proof of Proposition 2. Let $(u, v) \in V$. By Lemma 3,

$$\|\mathsf{D}_{\boxplus_n}(u, v)\|^2 = (u, v)^*\mathsf{D}_{\boxplus_n}\mathsf{D}_{\boxplus_n}^*(u, v) = \|u\|^2 + \|v\|^2 - \sum_{i=1}^{n} \gamma_i\,(u, v)^* H^{(i)}(u, v).$$

The roots $\gamma_1 \geq \cdots \geq \gamma_n$ are ordered, so Corollary 1 (with $c_i = \gamma_i$) gives $\sum_i \gamma_i H^{(i)} \succeq 0$, whence $\sum_i \gamma_i (u, v)^* H^{(i)}(u, v) \geq 0$. The claim follows. □

Step 3: Blachman's argument

Proof of the Theorem. First, $\mathsf{S}_n^{\mathrm{ff}}(\alpha) \perp \mathbf{1}_n$: indeed $\sum_i \sum_{j\neq i}(\alpha_i - \alpha_j)^{-1} = 0$ since each unordered pair contributes $+$ and $-$ once. Likewise $\mathsf{S}_n^{\mathrm{ff}}(\beta) \perp \mathbf{1}_n$. So Proposition 2 applies to $(a\,\mathsf{S}_n^{\mathrm{ff}}(\alpha),\, b\,\mathsf{S}_n^{\mathrm{ff}}(\beta))$:

$$\left\|\mathsf{D}_{\boxplus_n}\left(a\,\mathsf{S}_n^{\mathrm{ff}}(\alpha), b\,\mathsf{S}_n^{\mathrm{ff}}(\beta)\right)\right\|^2 \;\le\; a^2\|\mathsf{S}_n^{\mathrm{ff}}(\alpha)\|^2 + b^2\|\mathsf{S}_n^{\mathrm{ff}}(\beta)\|^2.$$

Combining with Observation 1,

$$(a+b)^2\,\mathcal{K}_n^{\mathrm{ff}}(p\boxplus_n q) \;\le\; a^2\,\mathcal{K}_n^{\mathrm{ff}}(p) + b^2\,\mathcal{K}_n^{\mathrm{ff}}(q),$$

valid for all $a, b \ge 0$. Choosing $a = 1/\mathcal{K}_n^{\mathrm{ff}}(p)$ and $b = 1/\mathcal{K}_n^{\mathrm{ff}}(q)$, the right side becomes $1/\mathcal{K}_n^{\mathrm{ff}}(p) + 1/\mathcal{K}_n^{\mathrm{ff}}(q)$ and the factor $(a+b)^2$ becomes $\left(1/\mathcal{K}_n^{\mathrm{ff}}(p) + 1/\mathcal{K}_n^{\mathrm{ff}}(q)\right)^2$, so

$$\left(\tfrac{1}{\mathcal{K}_n^{\mathrm{ff}}(p)} + \tfrac{1}{\mathcal{K}_n^{\mathrm{ff}}(q)}\right)^2 \mathcal{K}_n^{\mathrm{ff}}(p\boxplus_n q) \;\le\; \tfrac{1}{\mathcal{K}_n^{\mathrm{ff}}(p)} + \tfrac{1}{\mathcal{K}_n^{\mathrm{ff}}(q)}.$$

Dividing through yields $\dfrac{1}{\mathcal{K}_n^{\mathrm{ff}}(p\boxplus_n q)} \ge \dfrac{1}{\mathcal{K}_n^{\mathrm{ff}}(p)} + \dfrac{1}{\mathcal{K}_n^{\mathrm{ff}}(q)}$. The simple-root reduction (Lemma 2) extends this to all real-rooted p, q. $\square$

Remarks

The proof mirrors Blachman's [4] derivation of the classical Stam inequality [6]: there the score of a sum is a conditional expectation of the marginal score and the inequality reduces to Cauchy–Schwarz. Here the role of that conditional-expectation step is played by Observation 1 (the score of $p\boxplus_n q$ is the image of the marginal scores under the Jacobian $\mathsf{D}_{\boxplus_n}$) together with the operator-norm bound Proposition 2. Real-rootedness of $\boxplus_n$ is used exactly once and essentially — through the hyperbolicity of (10) and the BGLS convexity theorem — which is why the inequality fails for convolutions that do not preserve real-rootedness.

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Problem 5

Solution to Problem 5: The $\mathcal{O}$-Slice Filtration for N_∞ Operads

Setup: transfer systems and indexing systems

Fix a finite group G. Following Balchin–Barnes–Roitzheim [4] and Rubin [5], the homotopy theory of N_∞ operads on G is governed by the combinatorics of *transfer systems*.

Definition 1 (transfer system)**.** A *transfer system* on G is a partial order, denoted $\to$, on the set $\mathrm{Sub}(G)$ of subgroups of G satisfying:

1. it refines subgroup inclusion: if $H \to K$ then $H \subseteq K$;
2. it is conjugation invariant: if $H \to K$ and $g \in G$ then $gHg^{-1} \to gKg^{-1}$;
3. it is closed under restriction: if $H \to K$ and $J \subseteq K$ then $H \cap J \to J$.

Transfer systems on G form a poset under refinement.

Definition 2 (admissible sets, indexing system)**.** Let $\mathcal{O}$ be a transfer system on G. A finite H-set $T = \coprod_i H/K_i$ is *admissible* for $\mathcal{O}$ if $K_i \to H$ for all i. The collection $\mathcal{O}(H)$ of admissible H-sets, as H varies, is the associated *indexing system*; we also denote the corresponding N_∞ operad by $\mathcal{O}$.

For a finite G-set T we have the T*-norm* $N^T : \mathrm{Sp}^G \to \mathrm{Sp}^G$, built from the Hill–Hopkins–Ravenel norms [1] by $N^{G/H}(E) = N_H^G i_H^*(E)$ and $N^{T_0 \sqcup T_1}(E) = N^{T_0}(E) \otimes N^{T_1}(E)$.

The $\mathcal{O}$-slice filtration

We use equivariant localization (nullification) to construct the relative slice tower. Recall that the equivariant localizing subcategory generated by a set of objects is its closure under homotopy colimits, retracts, and tensors with orbit spectra.

Definition 3 ($\mathcal{O}$-slice connective spectra)**.** For an indexing system $\mathcal{O}$, let $\tau_{\geq n}^{\mathcal{O}}$ be the equivariant localizing subcategory of Sp^G generated by

$$\left\{ G_+ \otimes_H N^T S^1 \;\middle|\; T \in \mathcal{O}(H),\ |T| \geq n \right\}.$$

This is the category of $\mathcal{O}$*-slice* n*-connective* spectra.

Remark 1. There is an equivariant homeomorphism $N^T S^1 \cong S^{\mathbb{R}\cdot T}$, the representation sphere of the permutation representation of T. So the n-connective spectra are generated by representation spheres of admissible sets of cardinality $\geq n$. This replaces the regular-representation slice cells $G_+ \wedge_H S^{m\rho_H}$ of the complete (Hill–Hopkins–Ravenel) filtration: in the incomplete setting one is only permitted the norms indexed by sets that $\mathcal{O}$ actually admits.

Definition 4. For an indexing system $\mathcal{O}$ and $n \geq 0$:

- the nth $\mathcal{O}$*-slice truncation* $P_{\mathcal{O}}^n : \mathrm{Sp}_{\geq 0}^G \to \mathrm{Sp}_{\geq 0}^G$ is the nullification killing $\tau_{\geq (n+1)}^{\mathcal{O}}$;
- the nth $\mathcal{O}$*-slice cover* $P_n^{\mathcal{O}}$ is the fiber of $\mathrm{Id} \Rightarrow P_{\mathcal{O}}^{n-1}$;
- the nth $\mathcal{O}$*-slice* $P_{n,\mathcal{O}}^n(E)$ is the fiber of $P_{\mathcal{O}}^n(E) \to P_{\mathcal{O}}^{n-1}(E)$.

A G-spectrum E is an *$\mathcal{O}$-n-slice* if $E \in \tau^{\mathcal{O}}_{\geq n}$ and $E \to P^n_{\mathcal{O}} E$ is an equivalence.

That the tower converges (for E connective, $E \xrightarrow{\sim} \varprojlim P^n_{\mathcal{O}} E$) follows because the Postnikov connectivity of $G_+ \otimes_H N^T S^1$ is $|T/H| \to \infty$ with $|T|$. The suspension $\Sigma N^T S^1 \simeq N^{T \sqcup *} S^1$ shows $\Sigma : \tau^{\mathcal{O}}_{\geq k} \to \tau^{\mathcal{O}}_{\geq (k+1)}$, whence the ∞-category of $\mathcal{O}$-k-slices is discrete.

Geometric fixed points and the characteristic function

Stable equivalences in Sp^G are detected on geometric fixed points Φ^H for all $H \subseteq G$, so the connectivity of geometric fixed points is the governing invariant.

Definition 5 (geometric connectivity)**.** For $E \in \mathrm{Sp}^G$, let $\mathrm{gconn}(E) : \mathrm{Sub}(G) \to \mathbb{Z} \cup \{\pm\infty\}$ be $\mathrm{gconn}(E)(H) := \mathrm{conn}(\Phi^H E)$.

Definition 6 (characteristic function)**.** For a transfer system $\mathcal{O}$, define $\chi^{\mathcal{O}} : \mathrm{Sub}(G) \to \mathrm{Sub}(G)$ by

$$\chi^{\mathcal{O}}(H) \;:=\; \min\{K \mid K \to H\} \;=\; \bigcap_{K \to H} K.$$

The index $[G : \chi^{\mathcal{O}}(G)]$ is the maximal cardinality of an admissible G-set with a single orbit; equivalently $[G : \chi^{\mathcal{O}}(H)]$ measures how large an admissible H-set $\mathcal{O}$ permits per orbit. This subgroup, *not* $|H|$ itself, is what enters the connectivity bound.

Main Theorem

Theorem 1. *Let $\mathcal{O}$ be a transfer system on G and let E be a connective G-spectrum. Then $E \in \tau^{\mathcal{O}}_{\geq n}$ if and only if for all $H \subseteq G$,*

$$[H : \chi^{\mathcal{O}}(H)] \cdot \mathrm{gconn}(E)(H) \;\geq\; n. \tag{11}$$

Equivalently, writing $n = \min_{H \subseteq G} [H : \chi^{\mathcal{O}}(H)] \cdot \mathrm{gconn}(E)(H)$, one has $E \in \tau^{\mathcal{O}}_{\geq n}$; this n is the *$\mathcal{O}$-slice connectivity* of E.

When $\mathcal{O}$ is complete, $\chi^{\mathcal{O}}(H) = \{e\}$ for all H, $[H : \chi^{\mathcal{O}}(H)] = |H|$, and (11) recovers the Hill–Yarnall reformulation [2] of the Hill–Hopkins–Ravenel slice connectivity.

Proof

Forward direction (necessity)

Lemma 1. *If $E \in \tau^{\mathcal{O}}_{\geq n}$ then $[H : \chi^{\mathcal{O}}(H)] \cdot \mathrm{gconn}(E)(H) \geq n$ for all $H \subseteq G$.*

Proof. By restriction it suffices to treat $H = G$. Geometric fixed points preserve homotopy colimits and extensions and annihilate induced spectra from proper subgroups, so it suffices to check the generators $N^T S^1$ for admissible G-sets T with $|T| \geq n$. Decompose $T = \sum_H n_H \, G/H$. Then $\Phi^G(N^T S^1) = S^{|T/G|}$ with $|T/G| = \sum_H n_H$, while

$$|T| = \sum_H n_H [G : H] \geq n.$$

By definition $[G:\chi^{\mathcal{O}}(G)]$ is the maximal $[G:H]$ with $G/H \in \mathcal{O}(*)$ (and all others divide it), so

$$[G:\chi^{\mathcal{O}}(G)] \cdot \sum_H n_H \;\geq\; \sum_H n_H [G:H] \;\geq\; n,$$

i.e. $[G:\chi^{\mathcal{O}}(G)] \cdot \mathrm{gconn}(N^T S^1)(G) \geq n$, as required. When $\chi^{\mathcal{O}}(G) = \{e\}$ this is exactly [2, Thm. 2.5]. $\square$

Converse via isotropy separation

Let $\mathcal{P}$ be the family of proper subgroups of G and consider the isotropy separation sequence

$$E\mathcal{P}_+ \otimes E \;\longrightarrow\; E \;\longrightarrow\; \widetilde{E\mathcal{P}} \otimes E.$$

We argue by downward induction on the subgroup lattice. The first piece is built from inductions $G/H_+ \otimes E$ for $H \in \mathcal{P}$, controlled by the inductive hypothesis on restrictions; the second piece is *geometric* and is handled by the following.

Lemma 2 (geometric slices)**.** *Let M be a geometric Mackey functor. For any $\mathcal{O}$, $\Sigma^k HM$ is a $k \cdot [G:\chi^{\mathcal{O}}(G)]$-$\mathcal{O}$-slice.*

Proof. Since M is geometric, for any finite G-set T the inclusion of fixed points $S^{|T/G|} \hookrightarrow N^T S^1$ induces an equivalence $S^{|T/G|} \otimes HM \xrightarrow{\sim} N^T S^1 \otimes HM$. For the lower bound, choose an admissible T with $|T|$ maximal subject to $|T/G| = k$, namely $T = k\, G/\chi^{\mathcal{O}}(G)$ (as $\chi^{\mathcal{O}}(G)$ is the minimal subgroup with $\chi^{\mathcal{O}}(G) \to G$); this gives $\Sigma^k HM \in \tau^{\mathcal{O}}_{\geq k[G:\chi^{\mathcal{O}}(G)]}$. For the upper bound, if T is admissible with $|T| > k[G:\chi^{\mathcal{O}}(G)]$ then $|T/G| > k$, so

$$[N^T S^1, \Sigma^k HM]^G \cong [\Phi^G N^T S^1, \Sigma^k HM(G/G)] \cong [S^{|T/G|}, \Sigma^k HM(G/G)] = 0.$$

Hence $\Sigma^k HM$ is exactly a $k[G:\chi^{\mathcal{O}}(G)]$-slice. $\square$

Lemma 3. *Let $\mathcal{F}$ be a family and τ an equivariant localizing subcategory. If $i_H^* E \in i_H^* \tau$ for all $H \in \mathcal{F}$, then $E\mathcal{F}_+ \otimes E \in \tau$.*

Proof. $E\mathcal{F}_+ \otimes E$ lies in the localizing subcategory generated by $G/H_+ \otimes E \cong G_+ \otimes_H i_H^* E$ for $H \in \mathcal{F}$; by hypothesis each such object is in τ. (Same argument as [2, Lem. 2.4].) $\square$

Proof of Theorem 1, converse. Suppose E is connective with $[H:\chi^{\mathcal{O}}(H)] \cdot \mathrm{gconn}(E)(H) \geq n$ for all H. In the isotropy separation sequence, Lemma 2 (applied to the homotopy Mackey functors of the geometric part $\widetilde{E\mathcal{P}} \otimes E$) shows its $\mathcal{O}$-slice connectivity is $\geq n$. By downward induction on the subgroup lattice, the hypothesis on proper subgroups together with Lemma 3 shows $E\mathcal{P}_+ \otimes E$ also has $\mathcal{O}$-slice connectivity $\geq n$. Since localizing subcategories are closed under extensions, $E \in \tau^{\mathcal{O}}_{\geq n}$. $\square$

This is precisely the $\mathcal{O}$-relative form of [2, Thm. 2.5]; the only new ingredient is that the multiplier $|H|$ is replaced by $[H:\chi^{\mathcal{O}}(H)]$, reflecting that $\mathcal{O}$ admits norms only up to index $[H:\chi^{\mathcal{O}}(H)]$ per orbit.

Examples and consistency checks

Complete $\mathcal{O}$. Here every G/H is admissible, $\chi^{\mathcal{O}}(H) = \{e\}$, and (11) reads $|H| \cdot \mathrm{gconn}(E)(H) \geq n$ for all H — the Hill–Hopkins–Ravenel slice connectivity in the Hill–Yarnall formulation [2, 1].

Trivial $\mathcal{O}$. No nontrivial transfers: $\chi^{\mathcal{O}}(H) = H$ for all H, so $[H : \chi^{\mathcal{O}}(H)] = 1$ and (11) becomes $\mathrm{gconn}(E)(H) \geq n$ for all H. The only admissible sets are trivial orbits, and the filtration is the geometric Postnikov tower.

$G = \mathbb{Z}/p$. The two transfer systems are trivial and complete. For the complete one, $\chi^{\mathcal{O}}(\mathbb{Z}/p) = \{e\}$ gives the bound $p \cdot \mathrm{gconn}(E)(\mathbb{Z}/p) \geq n$ at the top together with $\mathrm{gconn}(E)(e) \geq n$; for the trivial one both indices are 1 and the conditions are $\mathrm{gconn}(E)(H) \geq n$. These agree with the C_p-slice computations of [2].

Scope

The model-categorical and ∞-categorical foundations for the norm functors and equivariant localization are taken from [1, 4, 5]; the geometric-Mackey-functor input (Lemma 2) adapts [3, §6]. The argument is the Hill–Yarnall template [2] carried out with the indexing-system-restricted norms in place of the full regular-representation cells, and the characteristic function $\chi^{\mathcal{O}}$ tracking the available norms.

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Problem 6

ε-Light Subsets in Graphs

Statement and Answer

Let $G = (V, E, w)$ be a weighted graph on $n = |V|$ vertices with Laplacian $L = \sum_{(s,t)\in E} w(s,t)\,(\delta_s - \delta_t)(\delta_s - \delta_t)^\top$. For $S \subseteq V$ let $G_S = (V, E(S,S))$ keep only edges with both endpoints in S and let L_S be its Laplacian (same ambient dimension as L). S is *ε-light* iff $\varepsilon L - L_S \succeq 0$.

Theorem. *For every weighted graph $G = (V, E, w)$ and every $\varepsilon \in (0, 1)$, there exists an ε-light subset $S \subseteq V$ with $|S| \geq \lfloor \varepsilon n/42 \rfloor$.*

The answer is **YES**, with the explicit constant $c = 1/42$.

Notation and basic facts

Write $L^\dagger$ for the Moore–Penrose pseudoinverse and $L^{\dagger/2}$ for its PSD square root. Define

$$\widetilde{L}_S := L^{\dagger/2} L_S L^{\dagger/2}, \qquad \widetilde{L}_{S,T} := L^{\dagger/2} L_{S,T} L^{\dagger/2}, \qquad \widetilde{L}_{S,t} := L^{\dagger/2} L_{S,\{t\}} L^{\dagger/2}.$$

Then $L^{\dagger/2} L L^{\dagger/2} = \Pi$, the orthogonal projector onto $\operatorname{span}\{L\}$, and the condition $\varepsilon L \succeq L_S$ is equivalent (by conjugation with $L^{\dagger/2}$ on $\operatorname{im} L$) to the operator-norm bound

$$\left\| \widetilde{L}_S \right\|_{\mathrm{op}} \;\leq\; \varepsilon. \tag{12}$$

Leverage scores. The leverage score of an edge (s,t) is

$$\ell(s,t) := w(s,t)\,(\delta_s - \delta_t)^\top L^\dagger (\delta_s - \delta_t) = \operatorname{Tr}\bigl(L_{\{s\},\{t\}} L^\dagger\bigr),$$

and zero on non-edges. For $S, T \subseteq V$ set $\ell(S,T) := \sum_{s\in S, t\in T} \ell(s,t)$ and write $\ell(S) := \ell(S, V\setminus S)$ for the boundary leverage. By linearity $\ell(S,T) = \operatorname{Tr}(\widetilde{L}_{S,T})$. Foster's identity, in the form $\sum_{(s,t)\in E} \ell(s,t) = n-1$ for connected G, follows from $\sum_{(s,t)} L_{\{s\},\{t\}} = L$ and $\operatorname{Tr}(LL^\dagger) = \operatorname{rank} L = n-1$.

Bounded-rank upper barrier. For a symmetric A with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$ and a real u, define

$$\mathcal{B}^u_\sigma(A) \;:=\; \begin{cases} \displaystyle\sum_{i=1}^{\sigma} \frac{1}{u - \lambda_i} & \text{if } u > \lambda_1, \\ \infty & \text{otherwise,} \end{cases}$$

and the top-σ trace $\operatorname{Tr}_\sigma(A) := \sum_{i=1}^{\sigma} \lambda_i$. We write $\mathcal{B}^u_\sigma(S)$ for $\mathcal{B}^u_\sigma(\widetilde{L}_S)$, and note $\mathcal{B}^u_\sigma(A) = \operatorname{Tr}_\sigma\bigl((uI - A)^{-1}\bigr)$ when $u > \lambda_1$.

For the rest of this note we fix

$$\delta := \frac{21}{n}, \qquad \phi := \frac{n}{21}, \qquad \sigma := \lfloor \varepsilon n/42 \rfloor.$$

Proof

We build S greedily, adding one vertex at a time and increasing the barrier parameter u by δ at each step. Throughout, two invariants are maintained: the bounded-rank barrier stays at most ϕ, and the boundary leverage stays at most $4|S|$.

Step 1: Initial set and target

Pick any $v_0 \in V$ and set $S_0 := \{v_0\}$, $u_0 := \varepsilon/2$. Since G_{S_0} has no edges, $\widetilde{L}_{S_0} = 0$, and

$$\mathcal{B}_\sigma^{u_0}(S_0) \;=\; \frac{\sigma}{u_0} \;\le\; \frac{\varepsilon n/42}{\varepsilon/2} \;=\; \frac{n}{21} \;=\; \phi.$$

After σ greedy steps, the barrier will sit at $u_\sigma = u_0 + \sigma\delta$. We chose parameters so that

$$u_\sigma = \tfrac{\varepsilon}{2} + \sigma \cdot \tfrac{21}{n} \le \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} = \varepsilon,$$

so once we reach $|S| = \sigma + 1$ with $\mathcal{B}_\sigma^{u_\sigma}(S)$ finite, every eigenvalue of $\widetilde{L}_S$ lies below ε and (12) gives $\varepsilon L \succeq L_S$.

Step 2: Greedy lemma

Lemma 1. *Suppose $|S| \le \sigma$, $\ell(S) \le 4|S|$, and $\mathcal{B}_\sigma^u(S) \le \phi$. Then there exists $t \notin S$ with*

$$\mathcal{B}_\sigma^{u+\delta}(S \cup \{t\}) \le \phi \qquad \textit{and} \qquad \ell(S \cup \{t\}) \le \ell(S) + 4.$$

The lemma is proved by two averaging arguments: more than half the candidates t keep the leverage budget (Step 3), and at least half keep the barrier (Step 4). The two sets therefore intersect.

Step 3: Leverage-score averaging

Claim 2. *For every $T \subseteq V$, $\sum_{t\in T} \ell(t,T) \le 2(|T|-1)$.*

Proof. By symmetry, $\sum_{t\in T} \ell(t,T) = 2\,\mathrm{Tr}(L_T L^\dagger)$. Since $L_T \preceq L$, all eigenvalues of $L_T L^\dagger$ lie in $[0,1]$, and L_T has rank at most $|T|-1$, so $\mathrm{Tr}(L_T L^\dagger) \le |T|-1$. □

For $t \notin S$,

$$\ell(S \cup \{t\}) \le \ell(S \cup \{t\}, V \setminus S) = \ell(S) + \ell(t, V \setminus S),$$

so it suffices to control $\ell(t, V\setminus S)$. Claim 2 applied to $T = V \setminus S$ gives $\sum_{t\notin S} \ell(t, V\setminus S) < 2|V \setminus S|$, whence by Markov more than half of $t \notin S$ satisfy $\ell(t, V\setminus S) \le 4$.

Step 4: Barrier averaging

This is the technical heart. The barrier-update lemma (Lemma 3 below) reduces our task to showing that for at least half of $t \notin S$,

$$U(S,t) \;:=\; \frac{\mathrm{Tr}_\sigma(M^{-2}\widetilde{L}_{S,t})}{\mathcal{B}_\sigma^u(S) - \mathcal{B}_\sigma^{u+\delta}(S)} + \mathrm{Tr}_\sigma(M^{-1}\widetilde{L}_{S,t}) \;<\; 1, \tag{13}$$

where $M := (u+\delta)I - \widetilde{L}_S$. We claim

$$\sum_{t\notin S} U(S,t) \;\le\; \tfrac{5}{\delta} + 5\phi. \tag{14}$$

Granting (14), since each $U(S,t) \geq 0$, more than half of $t \notin S$ satisfy

$$U(S,t) \leq \frac{2}{n-|S|}\left(\tfrac{5}{\delta}+5\phi\right) \leq \frac{2}{41n/42} \cdot \tfrac{10n}{21} < 1$$

for $|S| \leq \sigma \leq n/42$.

Proof of (14). Let Π_S project onto $\operatorname{im} \widetilde{L}_S$, and $\Pi_S^{\perp} := I - \Pi_S$. Since $M = (u+\delta)(\Pi_S + \Pi_S^{\perp}) - \widetilde{L}_S$ and $\Pi_S^{\perp}\widetilde{L}_S = 0$,

$$M^p \;=\; (u+\delta)^p \Pi_S^{\perp} + \big((u+\delta)\Pi_S - \widetilde{L}_S\big)^p, \qquad p \in \{-1,-2\}.$$

By Ky Fan subadditivity of Tr_σ [1, Thm. 4.3.47a],

$$\operatorname{Tr}_\sigma(M^p \widetilde{L}_{S,t}) \;\leq\; \operatorname{Tr}_\sigma\big((u+\delta)^p \Pi_S^{\perp}\widetilde{L}_{S,t}\big) + \operatorname{Tr}_\sigma\Big(((u+\delta)\Pi_S - \widetilde{L}_S)^p \widetilde{L}_{S,t}\Big). \tag{15}$$

$\Pi_S^{\perp}$ *piece.* $\operatorname{Tr}_\sigma(\Pi_S^{\perp}\widetilde{L}_{S,t}) \leq \operatorname{Tr}(\widetilde{L}_{S,t}) = \ell(S,t)$ (the PSD product has non-negative eigenvalues), and summing,

$$\sum_{t \notin S} \operatorname{Tr}_\sigma\big((u+\delta)^p \Pi_S^{\perp}\widetilde{L}_{S,t}\big) \leq (u+\delta)^p\, \ell(S) \leq (u+\delta)^p \cdot 4|S|.$$

Π_S *piece.* $\sum_{t \notin S} \widetilde{L}_{S,t} = \widetilde{L}_{S,V\setminus S} \preceq \Pi$, and $((u+\delta)\Pi_S - \widetilde{L}_S)^p$ is PSD on $\operatorname{im}\Pi_S$, so

$$\begin{aligned}\sum_{t \notin S} \operatorname{Tr}_\sigma\Big(((u+\delta)\Pi_S - \widetilde{L}_S)^p \widetilde{L}_{S,t}\Big) &\leq \operatorname{Tr}\Big(((u+\delta)\Pi_S - \widetilde{L}_S)^p \widetilde{L}_{S,V\setminus S}\Big) \\ &\leq \operatorname{Tr}\Big(((u+\delta)\Pi_S - \widetilde{L}_S)^p\Big) = \operatorname{Tr}(\Pi_S M^p \Pi_S) \leq \operatorname{Tr}_\sigma(M^p),\end{aligned}$$

the last step by the Ky Fan max principle.

Since all eigenvalues of M are at most $u+\delta$, for $p<0$ all eigenvalues of M^p are at least $(u+\delta)^p$, so $\operatorname{Tr}_\sigma(M^p) \geq \sigma(u+\delta)^p \geq |S|(u+\delta)^p$. Combining,

$$\sum_{t \notin S} \operatorname{Tr}_\sigma(M^p \widetilde{L}_{S,t}) \;\leq\; 4|S|(u+\delta)^p + \operatorname{Tr}_\sigma(M^p) \;\leq\; 5\ \operatorname{Tr}_\sigma(M^p).$$

Now we use (Claim 4) $\mathcal{B}_\sigma^u(S) - \mathcal{B}_\sigma^{u+\delta}(S) \geq \delta\ \operatorname{Tr}_\sigma(M^{-2})$ to bound the first term of $U(S,t)$, and $\operatorname{Tr}_\sigma(M^{-1}) = \mathcal{B}_\sigma^u(S) \leq \phi$ for the second:

$$\sum_{t \notin S} U(S,t) \;\leq\; \frac{5\ \operatorname{Tr}_\sigma(M^{-2})}{\delta\ \operatorname{Tr}_\sigma(M^{-2})} + 5\,\mathcal{B}_\sigma^u(S) \;\leq\; \tfrac{5}{\delta} + 5\phi,$$

which is (14). □

Step 5: Combining and conclusion

Iterating Lemma 1 σ times starting from S_0 yields S with $|S| = \sigma+1$, $\ell(S) \leq 4\sigma$, and $\mathcal{B}_\sigma^{u_\sigma}(S) < \infty$, hence (by Step 1) every eigenvalue of $\widetilde{L}_S$ is at most $u_\sigma \leq \varepsilon$, so $\varepsilon L \succeq L_S$. This proves the theorem with $|S| \geq \lfloor \varepsilon n/42 \rfloor$. □

Auxiliary lemmas

Lemma 3 (Barrier update)**.** *Let $A, B \succeq 0$ and $\delta > 0$. Set $M := (u+\delta)I - A$. If $\mathcal{B}_\sigma^u(A) < \infty$ and*

$$\frac{\operatorname{Tr}_\sigma(M^{-2}B)}{\mathcal{B}_\sigma^u(A) - \mathcal{B}_\sigma^{u+\delta}(A)} + \operatorname{Tr}_\sigma(M^{-1}B) \; < \; 1,$$

then $\mathcal{B}_\sigma^{u+\delta}(A+B) \le \mathcal{B}_\sigma^u(A)$.

Proof. $\mathcal{B}_\sigma^u(A) < \infty$ gives $u > \lambda_{\max}(A)$, hence $M, M^{-1}, M^{-2} \succ 0$. Write $B = CC^\top$. By Sherman–Morrison–Woodbury,

$$(M - CC^\top)^{-1} = M^{-1} + M^{-1}C(I - C^\top M^{-1}C)^{-1}C^\top M^{-1}.$$

Since $\operatorname{Tr}_\sigma(M^{-1}B) = \operatorname{Tr}_\sigma(C^\top M^{-1}C) < 1$ the inverse exists and the second summand is PSD, so $\lambda_{\max}(A+B) < u+\delta$. By Ky Fan subadditivity [1, Thm. 4.3.47a],

$$\mathcal{B}_\sigma^{u+\delta}(A+B) \le \operatorname{Tr}_\sigma(M^{-1}) + \operatorname{Tr}_\sigma\big((I - C^\top M^{-1}C)^{-1}C^\top M^{-2}C\big).$$

Since $\|C^\top M^{-1}C\| \le \operatorname{Tr}_\sigma(C^\top M^{-1}C) < 1$, the operator $(I - C^\top M^{-1}C)^{-1} \preceq (1 - \operatorname{Tr}_\sigma(C^\top M^{-1}C))^{-1} I$, and monotonicity of Tr_σ on PSD-ordered products gives

$$\operatorname{Tr}_\sigma\big((I - C^\top M^{-1}C)^{-1}C^\top M^{-2}C\big) \le \frac{\operatorname{Tr}_\sigma(M^{-2}B)}{1 - \operatorname{Tr}_\sigma(M^{-1}B)}.$$

Combining with $\operatorname{Tr}_\sigma(M^{-1}) = \mathcal{B}_\sigma^u(A) - (\mathcal{B}_\sigma^u(A) - \mathcal{B}_\sigma^{u+\delta}(A))$ and the hypothesis gives the claimed bound. ☐

Claim 4. $\mathcal{B}_\sigma^u(S) - \mathcal{B}_\sigma^{u+\delta}(S) \ge \delta \operatorname{Tr}_\sigma(M^{-2})$.

Proof. For each top-σ eigenvalue λ_i of $\widetilde{L}_S$,

$$\tfrac{1}{u-\lambda_i} - \tfrac{1}{u+\delta-\lambda_i} = \tfrac{\delta}{(u-\lambda_i)(u+\delta-\lambda_i)} \ge \tfrac{\delta}{(u+\delta-\lambda_i)^2};$$

sum over i. ☐

Remarks

1. **Constant.** The constant $1/42$ comes from the parameter choices $\delta = 21/n$, $\phi = n/21$, which balance the $5/\delta$ and 5ϕ contributions in (14). It is not believed optimal; improving it is open.
2. **Key innovation.** The standard upper-barrier function tracks all eigenvalues; here we track only the top σ. This adaptation, together with the Ky Fan inequalities for Tr_σ, is what makes the argument go through.
3. **Connection to spectral sparsification.** An ε-light S with $|S| \ge c\varepsilon n$ says the induced subgraph on S contributes at most εL spectrally, giving a perspective on "light" as "spectrally independent".
4. **Vertex vs. edge sampling.** The edge-partitioning analogue admits an optimal constant of $1/2$ via the Kadison–Singer resolution. The vertex-light variant is more delicate because edges sharing a vertex are correlated; the deterministic barrier method bypasses concentration entirely.

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Problem 7

Uniform Lattices with 2-Torsion as Fundamental Groups

Statement and Answer

Let Γ be a uniform (cocompact) lattice in a real semisimple Lie group G, and suppose Γ contains an element of order 2.

Theorem. *Such a Γ cannot be the fundamental group of a closed manifold without boundary whose universal cover is acyclic over $\mathbb{Q}$.*

The answer is **NO**.

A torsion-free uniform lattice in G *is* the fundamental group of such a manifold — a closed locally symmetric space, whose universal cover is the contractible symmetric space $X = G/K$. The content of the theorem is that 2-torsion obstructs this. Fowler, in his Ph.D. thesis, proved the corresponding statement when Γ has *odd* torsion (a proof appears in [3]); the present argument extends Fowler's theorem to the case of 2-torsion, following Cappell–Weinberger–Yan.

We stress at the outset what does *not* work: a proof using only that M is a finite Poincaré complex over $\mathbb{Q}$ cannot succeed, because such complexes with these fundamental groups exist (Proposition 1). The argument must use that M is a genuine manifold, and it does so through the symmetric signature and an equivariant fundamental-class computation.

What a finite-complex / Poincaré-duality argument cannot see

It is tempting to argue purely with rational Poincaré duality and Euler characteristics. That route is blocked by an explicit construction of Fowler.

Proposition 1. *There exist groups of the form $\pi_1(M_3) \times \Gamma$, with Γ a lattice with torsion, that are fundamental groups of spaces having the rational homotopy type of a finite complex satisfying rational Poincaré duality.*

Construction. Let Γ be a lattice in a linear semisimple group G, and choose a homomorphism $\Gamma \to \Delta$ to a finite group with torsion-free kernel Γ_0. Let $E\Delta$ be a contractible free Δ-space and M_3 any closed hyperbolic 3-manifold. The space

$$M_3 \times \big(K\backslash G/\Gamma_0 \times E\Delta\big)/\Delta$$

has the rational homotopy type of a finite complex and satisfies rational Poincaré duality; its fundamental group is $\pi_1(M_3) \times \Gamma$, a lattice in $\mathrm{SO}(3,1) \times G$. $\square$

In particular, "multiplicativity of the Euler characteristic in finite covers" *fails* for infinite complexes with finitely generated rational homology: a rationally acyclic example with rational Euler characteristic 1 is the universal cover of $\mathbb{R}P^2 \vee (\bigvee_\infty S^2)$ after equivariantly attaching cells $D^3 \times \mathbb{Z}/2$ along a free $\mathbb{Z}[1/2][\mathbb{Z}/2]$-basis of $\pi_2 = \mathbb{Z}[-1] \oplus \mathbb{Z}[\mathbb{Z}/2]^\infty$ to kill the homology; both it and its $\mathbb{Z}/2$-quotient become rationally acyclic with rational Euler characteristic 1. Hence the obstruction cannot be an Euler-characteristic denominator, and any proof relying on $\chi(M) \in \mathbb{Z}$ versus $\chi_{\mathbb{Q}}(\Gamma) \notin \mathbb{Z}$ is invalid. We also note that any proof resting on a naive Lefschetz fixed-point argument is unsound: the relevant Lefschetz-type statement is *false* (witness $\mathbb{R}^1$ with f a translation, which has no fixed points yet Lefschetz number -1 in that sense).

Proof of the Theorem

Reduction

Passing to a normal subgroup of finite index, it suffices to treat $\Gamma = \pi \rtimes \mathbb{Z}_2$ for a torsion-free π, a lattice in G, such that there is an involution on $M_0 := K\backslash G/\pi$ — by isometries of the locally symmetric metric — whose fixed set F is nonempty. (If F is disconnected one argues componentwise; we write the connected case.) Here M_0 is a closed aspherical manifold, a $K(\pi, 1)$.

Suppose, for contradiction, that X^m is a closed manifold with $\pi_1(X) \cong \Gamma$ and rationally acyclic universal cover. Let $Y \to X$ be the 2-fold cover corresponding to $\pi \leq \Gamma$; then $\pi_1(Y) \cong \pi$ and the universal cover of Y (the same as that of X) is rationally acyclic.

Step 1: a degree-one fundamental class

Work with symmetric signatures in the symmetric L-group $L(\mathbb{R}\pi)$ over the reals. (Over $\mathbb{R}$ the symmetric and quadratic L-groups agree, so we may write $L(\mathbb{R}\pi)$ unambiguously.) There is a map $f : Y \to M_0$ inducing the identity on $\pi_1 = \pi$. It need not have degree one, but over $\mathbb{R}$ every degree has a square root, so the symmetric signature is sensitive only to the sign of the degree, and f induces an equivalence of symmetric signatures.

Because the Novikov conjecture holds for π (a lattice in a semisimple group; it satisfies the Borel/Novikov package via its proper action on the nonpositively curved X), the assembly map

$$H_m\big(B\pi; \mathbb{L}(\mathbb{R})\big) \;\longrightarrow\; L_m(\mathbb{R}\pi)$$

is injective. The symmetric signature of Y is detected in the degree-m piece $H_m(B\pi; \mathbb{Z})$ as the image of the fundamental class. Injectivity forces

$$f_*[Y] \;=\; [M_0] \quad \text{in} \quad H_m(B\pi; \mathbb{Q}), \tag{16}$$

i.e. f is rationally degree one onto the aspherical M_0.

Step 2: an equivariant cobordism and the fixed set

We now run the cobordism argument of [2]. Consider any closed manifold Z with fundamental group π carrying an involution inducing the given automorphism of π, together with the image of $[Z]$ in $H_m(B\Gamma; \mathbb{Z}_2)$. Standard equivariant homotopy theory gives an equivariant map $g : Z \to M_0$, hence a map of fixed sets $Z^{\mathbb{Z}_2} \to F$. We claim

$$g_*[Z] \;=\; g_*[Z^{\mathbb{Z}_2}], \tag{17}$$

using the map $\mathbb{Z}_2 \times \pi_1 F \to \Gamma$ and the periodicity in the group homology of $\mathbb{Z}_2$ to raise the dimension from $\dim F$ to $\dim M_0$.

The relevant cobordism realizing (17) runs between Z and a projective-space bundle over $Z^{\mathbb{Z}_2}$ — the projectivized normal bundle of $Z^{\mathbb{Z}_2}$ — whose fundamental class is the asserted element by the Leray–Hirsch theorem. Explicitly it is $Z \times [0, 1]$, with $Z \times \{1\}$ modified by quotienting the free $\mathbb{Z}/2$-action on the complement of an equivariant regular neighbourhood of $Z^{\mathbb{Z}_2}$.

Step 3: the contradiction

Apply Step 2 in two ways.

For Y. The deck $\mathbb{Z}_2$-action on Y (equivalently the action with quotient X) is *free*, so $Y^{\mathbb{Z}_2} = \varnothing$ and the right-hand side of (17) is 0. Hence the corresponding image of the fundamental class is 0.

For M_0. The isometric $\mathbb{Z}_2$-action on M_0 has fixed set F which is aspherical, and the Borel construction shows that $\mathbb{Z}_2 \times F \to \Gamma$ induces an injection on homology in dimension $\dim(M_0/\mathbb{Z}_2)$ (an isomorphism in higher dimensions; see [1]). Since the fundamental class of an aspherical manifold is nontrivial in its group homology, the corresponding image for M_0 is *nonzero*.

But (16) identifies the Y-image with the M_0-image rationally: the degree-one map f carries the (zero) class for Y to the (nonzero) class for M_0. This is a contradiction. Therefore no such closed manifold X exists, and Γ is not the fundamental group of a closed manifold with rationally acyclic universal cover. $\square$

Remarks

1. **Where the manifold hypothesis enters.** The argument uses that X is a manifold twice: to form the symmetric signature and apply assembly injectivity (Step 1), and to run the equivariant normal-bundle cobordism (Step 2). Proposition 1 shows neither can be replaced by "finite Poincaré complex" alone; any proof using only finite complexes and Poincaré duality must fail.

2. **Odd torsion.** The corresponding statement for odd torsion is Fowler's theorem [3]; the present argument adapts the involution case ($p = 2$), where the fixed-point and normal-bundle analysis replaces the Euler-characteristic counting available in nicer settings.

3. **Role of Novikov.** Only injectivity of the rational assembly map for π is needed, which holds for lattices in semisimple groups via their proper actions on the nonpositively curved symmetric space.

4. **Honest accounting.** The delicate points are (16) – (17): identifying the symmetric signature with the fundamental class under assembly, and the periodicity/Leray–Hirsch computation that pushes the fixed-set class up to top dimension. These are the substance of the surgery-theoretic argument and are stated here at the level of their structure.

References

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Problem 8

Smoothing Polyhedral Lagrangian Surfaces

Statement and Answer

Equip $\mathbb{R}^4$ with coordinates (q_1, q_2, p_1, p_2) and the standard symplectic form $\omega = \mathrm{d}p_1 \wedge \mathrm{d}q_1 + \mathrm{d}p_2 \wedge \mathrm{d}q_2$. A *polyhedral Lagrangian surface* $K \subset \mathbb{R}^4$ is a finite polyhedral complex, all of whose faces are Lagrangian, that is a topological submanifold of $\mathbb{R}^4$.

Theorem. *If K is a polyhedral Lagrangian surface in which exactly 4 faces meet at every vertex, then K admits a Lagrangian smoothing: a Hamiltonian isotopy K_t of smooth Lagrangian submanifolds parametrized by $t \in (0, 1]$, extending to a topological isotopy on $[0, 1]$ with $K_0 = K$.*

The answer is **YES**.

The proof has a local part (a normal form near each vertex and edge, from which a smooth structure on K is read off) and a global part (a fibration of Lagrangian planes transverse to K, with respect to which all sufficiently C^1-small "smoothing functions" yield genuine graphical Lagrangians). The 4-faces hypothesis enters in the local normal form (Lemma 1), where four Lagrangian quadrants force a standard symplectic basis; it is not a Maslov/parity condition.

1. Local model at a vertex

Lemma 1. *Let $\mathbb{R}^2 \to \mathbb{R}^4$ be an embedding that is linear on each of the four quadrants, with Lagrangian image Σ, and assume Σ is not contained in a single plane. Then there is a linear symplectic transformation of $\mathbb{R}^4$ carrying Σ to the product of the union of the positive coordinate axes in $\mathbb{R}^2_{p_1q_1}$ with that in $\mathbb{R}^2_{p_2q_2}$.*

Proof. Let (v_1, v_2, u_1, u_2) be the tangent vectors at the origin to the four edges of Σ, ordered so that cyclically adjacent vectors span the faces. The pairings $\omega(v_i, u_i)$ cannot vanish: if some $\omega(v_i, u_i) = 0$, then ω would vanish identically on a 3-dimensional subspace of $\mathbb{R}^4$, which is impossible for a symplectic form. After swapping $v_i \leftrightarrow u_i$ if needed we may take both pairings positive, and after rescaling we may take $\omega(v_1, u_1) = \omega(v_2, u_2) = 1$ with the cross pairings vanishing (the faces are Lagrangian). Then (v_1, v_2, u_1, u_2) is a standard symplectic basis, and the linear map $\partial_{p_i} \mapsto v_i$, $\partial_{q_i} \mapsto u_i$ is the required symplectomorphism, carrying Σ to the stated product of positive-axis unions. □

In the plane $\mathbb{R}^2_{pq}$ the symplectic pairing projects the union of the two positive axes homeomorphically onto the dual of the antidiagonal line $p = q$. Taking the product and applying Lemma 1:

Corollary 1. *There is a linear Lagrangian plane $L \subset \mathbb{R}^4$ such that the symplectic pairing $\mathbb{R}^4 \to L^\vee$ restricts to a homeomorphism $\Sigma \to L^\vee$.*

This equips Σ with a smooth structure (pulled back from $L^\vee$), which we fix henceforth. Given L, call a Lagrangian $\Lambda \subset \mathbb{R}^4$ *graphical* if the pairing restricts to a diffeomorphism $\Lambda \cong L^\vee$.

2. Smoothing functions

Choose a Lagrangian splitting $\mathbb{R}^4 \cong L \oplus L^\vee$. Each of the four quadrant-faces of Σ is graphical over $L^\vee$, hence the graph of the differential of a quadratic form; collecting these gives quadratic forms $\{q_{ij}\}_{i,j\in\pm}$ on $L^\vee$, agreeing to first order along the edges. Via $\Sigma \cong L^\vee$ (Corollary 1) write q_Σ for the resulting C^1 function on Σ that is q_{ij} on each face.

Definition 1. The space $\mathscr{S}(\Sigma)$ of *smoothing functions* is the set of C^1 functions $f : \Sigma \to \mathbb{R}$ such that $f + q_\Sigma$ is C^∞.

$\mathscr{S}(\Sigma)$ is invariant under adding smooth functions, and depends only on L, not on the splitting (a different splitting changes q_{ij} by a global quadratic form, i.e. a smooth function). To a smoothing function f we associate the Lagrangian $\Lambda_{\mathrm{d}f} \subset \mathbb{R}^4$ that is, piecewise, the graph of $\mathrm{d}f$ over each face — equivalently the graph of $\mathrm{d}(f + q_\Sigma)$ in the splitting.

Lemma 2. *$f \mapsto \Lambda_{\mathrm{d}f}$ is a bijection between graphical Lagrangians and smoothing functions on Σ modulo additive constants.*

Proof. In the splitting, $\Lambda_{\mathrm{d}f}$ is the graph of $\mathrm{d}(f + q_\Sigma)$ as a function on $L^\vee$; graphical Lagrangians over $L^\vee$ are exactly graphs of differentials of smooth functions, and $f + q_\Sigma$ is smooth precisely when $f \in \mathscr{S}(\Sigma)$. The construction does not use the splitting, so the bijection depends only on L. □

The same statements hold verbatim for an edge: a pair of Lagrangian half-planes meeting along a line, with the analogue of Corollary 1 provided by the next lemma.

Lemma 3. *If Σ is a pair of linear Lagrangian half-planes meeting along a line ℓ, the space of Lagrangian planes L for which the pairing $\Sigma \to L^\vee$ is a homeomorphism is contractible.*

Proof. Up to affine linear symplectomorphism, Σ is the product of the real axis in one $\mathbb{R}^2$-factor with the union of the positive axes in another. The pairing $\Sigma \to L^\vee$ is a homeomorphism iff L is transverse to both half-planes, i.e. the symplectic reduction of L along ℓ is a line in $\ell^\perp/\ell \cong \mathbb{R}^2$ meeting the interior of the positive quadrant — a contractible condition. The space of Lagrangian lifts of such a line is itself contractible (two lifts differ by the graph of a map $\ell' \to \ell$), giving the claim. □

3. Existence of small smoothing functions

The crux is that $\mathscr{S}(\Sigma)$ is *not* invariant under rescaling, so the existence of small smoothing functions is not automatic.

Lemma 4. *$\mathscr{S}(\Sigma)$ contains functions of arbitrarily small C^1-norm.*

Proof. Fix a partition of unity $\sum_\sigma \chi_\sigma = 1$ on Σ indexed by the strata, with χ_σ supported near σ and $\equiv 1$ on σ away from its boundary, of bounded C^k norms. Let χ_σ^ε be the composition of χ_σ with dilation by $1/\varepsilon$; these are uniformly bounded with $\|\chi_\sigma^\varepsilon\|_{C^1} = O(1/\varepsilon)$.

Choose a Lagrangian plane Λ_σ containing each stratum σ and transverse to L, with associated smoothing function f_σ. The first-order agreement of the faces along shared edges means f_σ and $f_{\sigma'}$ agree to first order along $\sigma \cap \sigma'$. Set $f^\varepsilon := \sum_\sigma \chi_\sigma^\varepsilon f_\sigma$. The partition-of-unity structure makes f^ε a smoothing function, and although $\|\nabla \chi_\sigma^\varepsilon\|$ grows like $1/\varepsilon$, it is supported where $f_\sigma - f_{\sigma'} = O(\varepsilon^2)$; by the product rule the C^1-norm of f^ε is therefore $O(\varepsilon)$. □

4. Global construction: a conormal fibration

To globalize, we replace the single plane L by a smoothly varying family.

Definition 2. A *conormal fibration dual to* K is a smoothly varying family L_z of affine Lagrangian planes, $z \in K$, such that: (i) near each vertex the L_z are translates of a single plane as in Corollary 1; (ii) L_z varies smoothly along edges and along faces; and (iii) in a collar of each edge the planes in the normal direction are translates of those along the edge, with *matched* inward normal fields on the two adjacent faces (opposite vectors under the identification $T_z\sigma \cong L_z^\vee \cong T_z\sigma'$).

Lemma 5. *K admits a conormal fibration which near each vertex agrees with the local choice of Corollary 1.*

Proof. Lemma 3 extends the vertex choices over the edges (the space of admissible L_z along an edge is contractible). Choosing a normal vector field to one face along each edge determines matched normals on the other; extending to the interior of the faces is unobstructed because the space of Lagrangian planes transverse to a given one is contractible. □

The fibration determines an open neighbourhood νK of K in which the fibres L_z are disjoint, hence a projection $\nu K \to K$. A Lagrangian contained in νK is *graphical* (with respect to L_z) if its projection to K is a diffeomorphism. The local correspondence of Lemma 2 globalizes:

Lemma 6. *Every graphical Lagrangian with respect to $\{L_z\}$ is the graph of a smoothing function on K; and every smoothing function of sufficiently small differential defines a graphical Lagrangian.*

Proof. The correspondence is local on K. At a point z, a plane $L_z^\vee$ transverse to L_z stays transverse to nearby fibres, so a neighbourhood of z in νK is modelled on the conormal bundle of $L_z^\vee$ (Weinstein's tubular neighbourhood theorem), and Lagrangians there are graphs of closed 1-forms, i.e. differentials of smoothing functions. □

Lemma 7. *There exist smoothing functions for K of arbitrarily small C^1-norm.*

Proof. Take a partition of unity $\sum_\alpha \rho_\alpha = 1$ on K indexed by strata, with ρ_α supported in the open star of α. Lemma 4 gives smoothing functions f_α of arbitrarily small C^1-norm on each open star; then $\sum_\alpha \rho_\alpha f_\alpha$ is a global smoothing function of small C^1-norm. □

5. Assembly of the smoothing

Proof of the Theorem. By Lemma 5 fix a conormal fibration $\{L_z\}$ with disjoint fibres on a neighbourhood νK. By Lemma 7 choose a sequence of smoothing functions $f_i \to 0$ in C^1 with $\mathrm{d}f_i$ small enough that, by Lemma 6, each $\Lambda_{\mathrm{d}f_i} \subset \nu K$ is a smooth embedded graphical Lagrangian. These $\Lambda_{\mathrm{d}f_i}$ are all isotopic to K by a piecewise-smooth isotopy and converge to K in the Hausdorff topology as $i \to \infty$ (equivalently as the C^1-norm $\to 0$).

Because $\Lambda_{\mathrm{d}f_i}$ and $\Lambda_{\mathrm{d}f_{i+1}}$ are graphs of differentials of smooth functions over the common base K (via the fibration), the linear interpolation of those functions produces a smooth Hamiltonian path of graphical Lagrangians connecting them. Concatenating these paths and reparametrizing yields a Hamiltonian isotopy K_t ($t \in (0,1]$) of smooth Lagrangians with $K_t \to K$ as $t \to 0$, extending to a topological isotopy on $[0,1]$ with $K_0 = K$. This is the desired Lagrangian smoothing. □

Role of the 4-faces-per-vertex hypothesis

The hypothesis is used exactly once, in Lemma 1: four Lagrangian quadrants meeting at a vertex, with edge tangents (v_1, v_2, u_1, u_2), force $\omega(v_i, u_i) \neq 0$ and hence a standard symplectic basis, which is what makes the symplectic pairing $\Sigma \to L^\vee$ a homeomorphism (Corollary 1) and supplies the smooth structure used throughout. With a different number of faces the local model changes and this normal form is unavailable; the mechanism is the linear symplectic geometry of the vertex, not a Maslov-class parity condition.

Remarks

1. **Why smoothing functions, not Legendrian fillings.** The smoothing is produced directly as graphs over K via the conormal fibration, so no holomorphic-curve or contact-fillability input is needed; the only analytic ingredients are Weinstein's neighbourhood theorem and a partition-of-unity estimate.

2. **The rescaling subtlety.** $\mathscr{S}(\Sigma)$ is closed under adding smooth functions but not under scalar multiplication, so small smoothing functions must be built by hand (Lemma 4); this is the one place a genuine estimate is required.

3. **Matched normals.** Property (iii) of Definition 2 ensures the two smooth structures on K — from Corollary 1 and from the edge collars — agree, so the global graph construction is consistent across edges.

4. **Generalization.** The argument is local-to-global and uses only linear symplectic normal forms plus partition-of-unity gluing, so it adapts to other ambient symplectic manifolds once the vertex normal form is established.

References

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Problem 9

Algebraic Relations on Determinantal Tensors

Setup and Statement

For $n \geq 5$, let $A^{(1)}, \ldots, A^{(n)} \in \mathbb{R}^{3\times 4}$ be Zariski-generic. For each ordered 4-tuple $(\alpha, \beta, \gamma, \delta) \in [n]^4$ define $Q^{(\alpha\beta\gamma\delta)} \in \mathbb{R}^{3\times3\times3\times3}$ by

$$Q^{(\alpha\beta\gamma\delta)}_{ijk\ell} \;=\; \det\!\big[A^{(\alpha)}(i,:);\, A^{(\beta)}(j,:);\, A^{(\gamma)}(k,:);\, A^{(\delta)}(\ell,:)\big], \qquad 1 \leq i,j,k,\ell \leq 3,$$

the 4×4 determinant of the matrix formed by stacking one row from each of the four matrices.

We seek a polynomial map $\mathbf{F} : \mathbb{R}^{81n^4} \to \mathbb{R}^N$ such that:

1. $\mathbf{F}$ does not depend on $A^{(1)}, \ldots, A^{(n)}$;
2. the degrees of the coordinate functions of $\mathbf{F}$ do not depend on n;
3. for $\lambda \in \mathbb{R}^{n\times n\times n\times n}$ with $\lambda_{\alpha\beta\gamma\delta} \neq 0$ precisely on the non-identical tuples, $\mathbf{F}\big(\lambda_{\alpha\beta\gamma\delta} Q^{(\alpha\beta\gamma\delta)}\big) = 0$ *iff* there exist $u, v, w, x \in (\mathbb{R}^*)^n$ with $\lambda_{\alpha\beta\gamma\delta} = u_\alpha v_\beta w_\gamma x_\delta$ for all non-identical $(\alpha, \beta, \gamma, \delta)$.

Theorem. *Such a map exists.* Assemble the tensors into a single $3n\times 3n\times 3n\times 3n$ block tensor $\mathbf{Q}$ whose $(\alpha, \beta, \gamma, \delta)$ block is $Q^{(\alpha\beta\gamma\delta)}$, and let $\mathbf{F}$ send the input to the 5×5 minors of the four $3n\times 27n^3$ mode flattenings of $\mathbf{Q}$. Each coordinate of $\mathbf{F}$ is a 5×5 determinant, hence degree 5 independent of n, and $\mathbf{F}$ satisfies the rank-one characterization.

The answer is **YES**.

The key identity: a Tucker decomposition of Q

Index the modes of $\mathbf{Q}$ by $[n]\times[3]$, so that the $\big((\alpha,i),(\beta,j),(\gamma,k),(\delta,\ell)\big)$ entry of $\mathbf{Q}$ is $Q^{(\alpha\beta\gamma\delta)}_{ijk\ell}$.

Lemma 1. $\mathbf{Q}$ *admits the Tucker decomposition*

$$\mathbf{Q} \;=\; \mathbf{C} \times_1 \mathbf{A} \times_2 \mathbf{A} \times_3 \mathbf{A} \times_4 \mathbf{A}, \tag{18}$$

where $\mathbf{A} = [A^{(1)}; \ldots; A^{(n)}] \in \mathbb{R}^{3n\times 4}$ *is the vertical concatenation of the data matrices, and* $\mathbf{C} \in \mathbb{R}^{4\times4\times4\times4}$ *is the sign tensor*

$$C_{abcd} = \begin{cases} \operatorname{sgn}(abcd) & a,b,c,d \in [4] \text{ distinct}, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. For $(\alpha,i),(\beta,j),(\gamma,k),(\delta,\ell) \in [n]\times[3]$, the definition of the Tucker product gives

$$(\mathbf{C} \times_1 \mathbf{A} \times_2 \mathbf{A} \times_3 \mathbf{A} \times_4 \mathbf{A})_{(\alpha,i),(\beta,j),(\gamma,k),(\delta,\ell)} = \sum_{a,b,c,d\in[4]} C_{abcd}\, \mathbf{A}_{(\alpha,i),a}\mathbf{A}_{(\beta,j),b}\mathbf{A}_{(\gamma,k),c}\mathbf{A}_{(\delta,\ell),d}.$$

Because $\mathbf{A}$ is the vertical concatenation, $\mathbf{A}_{(\alpha,i),a} = A^{(\alpha)}_{ia}$, and the sum over distinct a,b,c,d weighted by $\operatorname{sgn}(abcd)$ is exactly the Leibniz expansion of the 4×4 determinant of the rows $A^{(\alpha)}(i,:), A^{(\beta)}(j,:), A^{(\gamma)}(k,:), A^{(\delta)}(\ell,:)$. This equals $Q^{(\alpha\beta\gamma\delta)}_{ijk\ell} = \mathbf{Q}_{(\alpha,i),(\beta,j),(\gamma,k),(\delta,\ell)}$. $\square$

Equation (18) shows the multilinear rank of $\mathbf{Q}$ is at most $(4,4,4,4)$: each mode-m flattening factors through the 4-column matrix $\mathbf{A}$, so its column space (resp. row space) has dimension at most 4. In particular **every** 5×5 **minor of every flattening of Q vanishes**, and the same holds for any tensor of multilinear rank $\le (4,4,4,4)$. This is why $\mathbf{F}$, the collection of 5×5 minors of the four flattenings, is the right object: it tests membership in the variety of multilinear rank $\le (4,4,4,4)$ tensors.

Definition of F

Given input tensors $\{T^{(\alpha\beta\gamma\delta)} \in \mathbb{R}^{3\times3\times3\times3}\}$, assemble the $3n\times 3n\times 3n\times 3n$ block tensor $\mathbf{T}$ with blocks $T^{(\alpha\beta\gamma\delta)}$. Let $\mathbf{T}_{(m)} \in \mathbb{R}^{3n\times 27n^3}$ be its mode-m flattening, $m=1,2,3,4$. Define

$$\mathbf{F}(\{T^{(\alpha\beta\gamma\delta)}\}) \;:=\; \big(\text{all } 5\times 5 \text{ minors of } \mathbf{T}_{(1)}, \mathbf{T}_{(2)}, \mathbf{T}_{(3)}, \mathbf{T}_{(4)}\big).$$

Each coordinate is a 5×5 determinant in the entries of the input, hence a degree-5 polynomial in those entries, independent of n; and $\mathbf{F}$ visibly does not reference $A^{(1)},\dots,A^{(n)}$. Properties (1) and (2) hold. We write $\lambda \odot_b \mathbf{Q}$ for the blockwise scaling whose $(\alpha,\beta,\gamma,\delta)$ block is $\lambda_{\alpha\beta\gamma\delta} Q^{(\alpha\beta\gamma\delta)}$. Property (3) is the content of the next two sections.

The "if" direction

Proposition 1. *If $\lambda_{\alpha\beta\gamma\delta} = u_\alpha v_\beta w_\gamma x_\delta$ on the non-identical tuples (and 0 on the diagonal $\alpha=\beta=\gamma=\delta$), then $\mathbf{F}(\lambda\odot_b \mathbf{Q}) = 0$.*

Proof. First, the diagonal blocks of $\mathbf{Q}$ vanish: $Q^{(\alpha\alpha\alpha\alpha)}_{ijk\ell}$ is a determinant with rows drawn from the single 3×4 matrix $A^{(\alpha)}$, and any 4×4 determinant of vectors lying in a 3-dimensional row space is 0. Hence the value of λ on the diagonal is irrelevant and $\lambda\odot_b\mathbf{Q} = (u\otimes v\otimes w\otimes x)\odot_b \mathbf{Q}$.

Blockwise scaling by a rank-one tensor with nonzero entries is the same as a Tucker multiplication by invertible diagonal matrices:

$$(u\otimes v\otimes w\otimes x)\odot_b \mathbf{Q} \;=\; \mathbf{Q}\times_1 D_u \times_2 D_v \times_3 D_w \times_4 D_x,$$

where $D_u \in \mathbb{R}^{3n\times 3n}$ is the diagonal matrix repeating each u_α three times (once per row of the α block), and similarly D_v, D_w, D_x. Multiplying (18) on each mode by an invertible matrix preserves the multilinear rank, so $\lambda\odot_b\mathbf{Q}$ still has multilinear rank $\le (4,4,4,4)$, and all its 5×5 flattening minors vanish. Thus $\mathbf{F}(\lambda\odot_b\mathbf{Q}) = 0$. $\square$

The "only if" direction

Assume λ has nonzero entries exactly off the diagonal and $\mathbf{F}(\lambda\odot_b\mathbf{Q}) = 0$, i.e. $\lambda\odot_b\mathbf{Q}$ has multilinear rank $\le (4,4,4,4)$. We show λ is rank-one off the diagonal.

Normalization

Replacing λ by its entrywise product with $\bar u\otimes\bar v\otimes\bar w\otimes\bar x$, where e.g. $\bar u_1 = 1$ and $\bar u_\alpha = \lambda^{-1}_{\alpha 111}$ for $\alpha\ge 2$ (and $\bar v,\bar w,\bar x$ defined symmetrically on the other modes), preserves both the multilinear rank of $\lambda\odot_b\mathbf{Q}$ and the property of being rank-one off-diagonal. Hence we may assume

$$\lambda_{\alpha 111} = \lambda_{1\beta 11} = \lambda_{11\gamma 1} = \lambda_{111\delta} = 1 \qquad \text{for all } \alpha,\beta,\gamma,\delta\in\{2,\dots,n\}. \tag{19}$$

We will show there is a constant $c \in \mathbb{R}^*$ with

$$\lambda_{\alpha\beta\gamma\delta} = c^{m-1}, \qquad m := \#\{\text{indices among } \alpha, \beta, \gamma, \delta \text{ that are } \neq 1\}, \tag{20}$$

for all non-identical tuples. Then $u = v = w = (1, c, \ldots, c)$ and $x = (c^{-1}, 1, \ldots, 1)$ give $\lambda_{\alpha\beta\gamma\delta} = u_\alpha v_\beta w_\gamma x_\delta$, establishing rank one.

Reading off 5×5 submatrix determinants

The multilinear-rank hypothesis says every 5×5 minor of every flattening of $\lambda \odot_b \mathbf{Q}$ vanishes. We extract three families of such minors. Write $(\lambda \odot_b \mathbf{Q})_{(1)} \in \mathbb{R}^{3n \times 27n^3}$ for the mode-1 flattening; rows are indexed by $(\alpha, i) \in [n] \times [3]$ and columns by $((\beta, j), (\gamma, k), (\delta, \ell))$.

Step 1 (two non-unit indices). Choose the 5×5 submatrix with rows $(1,1), (1,2), (1,3), (\alpha, 1), (\alpha, 2)$ and a fixed list of five columns chosen so that, viewed as a polynomial in the two variables $\lambda_{\alpha\alpha 11}$ and $\lambda_{\alpha 1\beta 1}$, the determinant has degree at most 2; it vanishes when $\lambda_{\alpha 1 \beta 1} = 0$, and also when $\lambda_{\alpha 1\beta 1} = \lambda_{\alpha\alpha 11}$ (in the latter case the submatrix becomes a submatrix of the unscaled $\mathbf{Q}_{(1)}$ after row operations, hence rank-deficient by Lemma 1). Therefore this determinant equals

$$s \cdot \lambda_{\alpha 1\beta 1}\big(\lambda_{\alpha 1\beta 1} - \lambda_{\alpha\alpha 11}\big),$$

with $s = s(A^{(1)}, A^{(\alpha)}, A^{(\beta)})$ a polynomial in the data. A single generic numerical instance (it suffices to take $\alpha = \beta$, the most specialized case) shows $s \not\equiv 0$, so $s \neq 0$ Zariski-generically. The minor vanishes, forcing $\lambda_{\alpha 1\beta 1} = \lambda_{\alpha\alpha 11}$. Applying the same argument to all modewise permutations π (using $\pi \cdot \mathbf{Q} = \operatorname{sgn}(\pi)\mathbf{Q}$, so the permuted flattening has the same rank bound) gives $\lambda_{\pi(\alpha 1\beta 1)} = \lambda_{\pi(\alpha\alpha 11)}$. Combining the specializations $\alpha = \beta$ and $\alpha \neq \beta$ yields a single constant

$$c := \lambda_{\pi(\alpha\beta 11)} \quad \text{for all } \alpha, \beta \in \{2, \ldots, n\} \text{ and all } \pi,$$

i.e. every λ-entry with exactly two non-unit indices equals c.

Step 2 (one non-unit index). Take the 5×5 submatrix with rows $(1,1), (1,2), (1,3), (\alpha, 1), (\alpha, 2)$ and columns chosen so the determinant, viewed as a polynomial in c and $\lambda_{\alpha\beta\gamma 1}$, has degree ≤ 3, vanishes when $c = 0$ (two rows become dependent), and vanishes when $c^2 = \lambda_{\alpha\beta\gamma 1}$ (the submatrix reduces to one of $\mathbf{Q}_{(1)}$ after row/column operations). Hence the determinant is a data-dependent scalar times $c\,(c^2 - \lambda_{\alpha\beta\gamma 1})$. Generic non-vanishing of the scalar (checked on one instance with $\alpha = \beta = \gamma$) and the rank hypothesis give $\lambda_{\alpha\beta\gamma 1} = c^2$, and by symmetry $\lambda_{\pi(\alpha\beta\gamma 1)} = c^2$ for all π. Thus every λ-entry with exactly one non-unit index equals c^2.

Step 3 (no non-unit index). Take the 5×5 submatrix with rows $(1,1), (1,2), (1,3), (\alpha, 1), (\alpha, 2)$ and columns producing a determinant equal to a data scalar times $c\,(c^3 - \lambda_{\alpha\beta\gamma\delta})$, where now $\alpha, \beta, \gamma, \delta \in \{2, \ldots, n\}$ are not all equal (the construction parallels Steps 1–2; the most specialized instance $\alpha = \beta = \gamma$ certifies non-vanishing). The rank hypothesis forces $\lambda_{\alpha\beta\gamma\delta} = c^3$, and by symmetry $\lambda_{\pi(\alpha\beta\gamma\delta)} = c^3$.

Steps 1–3 establish (20): a λ-entry with m non-unit indices equals c^{m-1} (using the normalization (19) for $m = 1$). As noted, this is exactly the rank-one factorization $\lambda_{\alpha\beta\gamma\delta} = u_\alpha v_\beta w_\gamma x_\delta$ with $u = v = w = (1, c, \ldots, c)$ and $x = (c^{-1}, 1, \ldots, 1)$. This proves the "only if" direction and completes the proof of the Theorem. □

Degree, dimension, and independence

Degree. Each coordinate of $\mathbf{F}$ is a 5×5 determinant in the entries of the input, so has degree 5, independent of n. ✓

Independence of the data. $\mathbf{F}$ is defined purely as minors of the flattenings of the input block tensor; it never refers to $A^{(1)}, \dots, A^{(n)}$. ✓

Number of coordinates. N is the number of 5×5 minors of four $3n \times 27n^3$ matrices, which grows polynomially in n; only the *degree* is required to be bounded, which it is. ✓

Remarks

1. **Why 5×5 and not 2×2.** The obstruction is multilinear rank $(4, 4, 4, 4)$: a flattening has rank ≤ 4, so the vanishing of all 5×5 minors is the exact algebraic condition. Smaller minors do not characterize the variety, and larger ones are redundant.

2. **Role of genericity.** Genericity of the $A^{(i)}$ enters only in the "only if" direction, to certify that the data-dependent scalars s in Steps 1–3 are nonzero; each is verified by exhibiting one numerical instance, after which Zariski-genericity does the rest.

3. **Diagonal blocks vanish.** $Q^{(\alpha\alpha\alpha\alpha)} = 0$ because four rows of a 3×4 matrix are linearly dependent; this is what lets the off-diagonal rank-one structure of λ be detected without the diagonal interfering.

4. **Fourth-order analogue.** The construction is the order-4 analogue of the corresponding third-order determinantal-variety result; the block-tensor flattening and the multilinear-rank bound are the common mechanism.

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Problem 10

Preconditioned CG for an RKHS-Constrained CP Subproblem

Setup recap

We solve the linear system

$$\left[(Z \otimes K)^\top S\, S^\top (Z \otimes K) + \lambda (I_r \otimes K) \right] \operatorname{vec}(W) \;=\; (I_r \otimes K) \operatorname{vec}(B), \tag{21}$$

where $K \in \mathbb{R}^{n\times n}$ is the PSD RKHS kernel matrix on mode k, $Z \in \mathbb{R}^{M\times r}$ is the Khatri–Rao product of the other factor matrices, $S \in \mathbb{R}^{N\times q}$ is the sampling/selection matrix, $B = TZ \in \mathbb{R}^{n\times r}$ is the MTTKRP, $\lambda > 0$ is a Tikhonov regularizer, and the unknown is $W \in \mathbb{R}^{n\times r}$. We have $n, r, d < q \ll N$ and $n \ll M$, with $N = \prod_i n_i$ and $M = N/n$. The objective is to avoid any $O(N)$ or $O(M)$ work.

Notation. Let $\Omega \subseteq [N]$, $|\Omega| = q$, index the observed entries of the vectorized mode-k unfolding; $S^\top x = x_\Omega$, $S^\top S = I_q$, and $SS^\top = \operatorname{diag}(\mathbf{1}_\Omega)$. A vectorized index $\ell \in [N]$ corresponds to a pair (i, m) with $i \in [n]$ (mode-k index) and $m \in [M]$ (multi-index of the other modes). Write the observations as triples (i_t, m_t, τ_t) for $t = 1, \ldots, q$.

The system matrix in (21),

$$H \;:=\; (Z \otimes K)^\top S S^\top (Z \otimes K) + \lambda (I_r \otimes K),$$

is symmetric and PSD (a Gram term plus λ times the PSD matrix $I_r \otimes K$), but it is only positive *semi*definite when K is, and even when $K \succ 0$ it can be poorly conditioned. Our plan is: (i) symmetrize and add a small shift ρ for definiteness; (ii) diagonalize K to expose a diagonal preconditioner; (iii) run PCG with matrix-free products built from three Kronecker-row identities.

1. Direct symmetric method (baseline)

Adding a shift $\rho > 0$ to ensure positive definiteness, the direct method forms

$$H_\rho \;=\; F^\top F + \lambda (I_r \otimes K) + \rho I_{rn}, \qquad F := S^\top (Z \otimes K) \in \mathbb{R}^{q \times rn},$$

and solves $H_\rho \operatorname{vec}(W) = \operatorname{vec}(KB)$ by Cholesky. Row t of F is the Kronecker product

$$F(t,:) \;=\; \widehat{Z}(t,:) \otimes \widehat{K}(t,:), \qquad \widehat{Z}(t,:) := Z(m_t,:), \quad \widehat{K}(t,:) := K(i_t,:),$$

so $F^\top F = \sum_t \big(\widehat{Z}(t,:) \otimes \widehat{K}(t,:)\big)^\top \big(\widehat{Z}(t,:) \otimes \widehat{K}(t,:)\big)$. Each rank-one update costs $O(n^2 r^2)$, giving $O(qn^2r^2)$ to form $F^\top F$, plus $O(n^3 r^3)$ for Cholesky.

Total (direct): $O(qrd + qn^2r^2 + n^3r^3)$; storage $O(n^2r^2)$.

Algorithm 1 Direct symmetric solve for the RKHS mode-k subproblem

Require: triples $\{(i_t, m_t, \tau_t)\}_{t=1}^q$, factors $\{A_i\}_{i\neq k}$, kernel K, regularizer λ, shift ρ.
Ensure: $W \in \mathbb{R}^{n\times r}$.
1: Build sampled Khatri–Rao rows $\widehat{Z}(t,:) \leftarrow \bigodot_{i\neq k} A_i(\mathrm{idx}_i(m_t),:)$ for $t = 1,\ldots,q$.
2: Build sampled kernel rows $\widehat{K}(t,:) \leftarrow K(i_t,:)$.
3: Form $F \in \mathbb{R}^{q\times rn}$ row-wise: $F(t,:) \leftarrow \widehat{Z}(t,:) \otimes \widehat{K}(t,:)$.
4: Assemble $H_\rho \leftarrow F^\top F + \lambda(I_r \otimes K) + \rho I_{rn}$.
5: Accumulate $B(i,:) \leftarrow \sum_{t:i_t=i} \tau_t\, \widehat{Z}(t,:)$.
6: Form RHS $g \leftarrow \mathrm{vec}(KB)$.
7: Solve $H_\rho\, \mathrm{vec}(W) = g$ by Cholesky (or symmetric LDL$^\top$).
8: **return** W.

2. Transforming the system

We exploit the eigendecomposition $K = UDU^\top$ (with U orthogonal, D diagonal) to factor $(I_r \otimes U)$ out of (21). Writing $\bar{W} := U^\top W$ and $\bar{B} := DU^\top B$,

$$\Big[\underbrace{(Z\otimes UD)^\top S}_{\bar{F}^\top}\,\underbrace{S^\top (Z\otimes UD)}_{\bar{F}} + \lambda(I_r\otimes D) + \rho I_{rn}\Big]\ \mathrm{vec}(\bar{W}) \ = \ \mathrm{vec}(\bar{B}). \tag{22}$$

We recover $W = U\bar{W}$ after solving. We cannot fold D into $\bar{W}$ because D may be singular (or indefinite after a shift), but $\lambda(I_r \otimes D)$ is now *diagonal*, which is the crucial gain: the regularization and the shift together contribute a known diagonal, and (§4) the diagonal of $\bar{F}^\top\bar{F}$ can be computed exactly and cheaply. Define

$$\bar{F} := S^\top (Z\otimes UD) \in \mathbb{R}^{q\times rn}, \qquad H'_\rho := \bar{F}^\top \bar{F} + \lambda(I_r\otimes D) + \rho I_{rn}.$$

Row t of $\bar{F}$ is again a Kronecker product of two rows,

$$\bar{F}(t,:) = \widehat{Z}(t,:) \otimes \widehat{H}(t,:), \qquad \widehat{H}(t,:) := H(i_t,:), \quad H := UD \in \mathbb{R}^{n\times n}, \tag{23}$$

so all three lemmas of §3 apply with $A = \widehat{Z}$ and $B = \widehat{H}$.

3. Three Kronecker-row identities

The matrix-free products and the preconditioner rest on the following identities. Let $A \in \mathbb{R}^{q\times r}$ and $B \in \mathbb{R}^{q\times n}$, and define $C \in \mathbb{R}^{q\times rn}$ row-wise by

$$C(t,:) \ = \ A(t,:) \otimes B(t,:), \qquad t = 1,\ldots,q. \tag{24}$$

Under the column-major vec convention, an index $\ell \in [rn]$ corresponds to (i,j) with $i \in [n]$, $j \in [r]$ via $\ell = i + (j-1)n$; then $C_{t\ell} = B_{ti}A_{tj}$ and $(\mathrm{vec}\,X)_\ell = X_{ij}$.

Lemma 1 (forward product)**.** *For $X \in \mathbb{R}^{n\times r}$ and $x = \mathrm{vec}(X)$,*

$$Cx \ = \ \big(A * (BX)\big)\,\mathbf{1}_r,$$

where $*$ is the elementwise product and $\mathbf{1}_r$ the all-ones vector.

Proof. $(Cx)_t = \sum_\ell C_{t\ell}x_\ell = \sum_{j=1}^r \sum_{i=1}^n B_{ti}X_{ij}A_{tj} = \sum_{j=1}^r (BX)_{tj}A_{tj}$, which is row t of $(A*BX)\mathbf{1}_r$. $\square$

Lemma 2 (adjoint product)**.** *For* $v \in \mathbb{R}^q$, $C^\top v = \mathrm{vec}\big(B^\top \operatorname{diag}(v)A\big)$.

Proof. With $\ell = i+(j-1)n$, $(C^\top v)_\ell = \sum_t C_{t\ell}v_t = \sum_t B_{ti}A_{tj}v_t = (B^\top \operatorname{diag}(v)A)_{ij} = (\mathrm{vec}(B^\top \operatorname{diag}(v)A))_\ell$. $\square$

Lemma 3 (diagonal of the Gram)**.** $\operatorname{diag}(C^\top C) = \mathrm{vec}\big((B*B)^\top(A*A)\big)$.

Proof. $(C^\top C)_{\ell\ell} = \sum_t C_{t\ell}^2 = \sum_t B_{ti}^2A_{tj}^2 = \big((B*B)^\top(A*A)\big)_{ij}$. $\square$

Lemma 1 computes Cx in $O(q(r+n))$; Lemma 2 computes $C^\top v$ in $O(q(r+n))$; Lemma 3 computes the full diagonal of $C^\top C$ in $O(q(r^2+n^2))$ — all without forming C, which would need $O(qrn)$ storage.

4. PCG: matrix-free products and preconditioner

We never form $\bar F$, Z, S, or H'_ρ. PCG needs the right-hand side, the product $w \mapsto H'_\rho w$, and a preconditioner.

4.1 The matrix–vector product

Apply the row factorization (23) with $A = \widehat Z$, $B = \widehat H$ where $\widehat H(t,:) = H(i_t,:)$, $H = UD$. For $x = \mathrm{vec}(X)$, $X \in \mathbb{R}^{n\times r}$:

- By Lemma 1, $\bar F x = (\widehat Z * (\widehat H X))\mathbf{1}_r \in \mathbb{R}^q$. Forming $\widehat H X$ uses only the q needed rows of HX; in practice compute $HX = U(DX) \in \mathbb{R}^{n\times r}$ once at $O(n^2 r)$ and gather rows i_t.
- By Lemma 2, $\bar F^\top(\bar F x) = \mathrm{vec}\big(\widehat H^\top \operatorname{diag}(\bar F x)\,\widehat Z\big)$, an $O(q(r+n))$ scatter.

Hence

$$H'_\rho x = \mathrm{vec}\Big(\widehat H^\top \operatorname{diag}\big((\widehat Z * \widehat H X)\mathbf{1}_r\big)\,\widehat Z\Big) + \lambda\, DX + \rho X, \tag{25}$$

computed in $O(n^2 r + q(r+n))$ per iteration (the $\lambda(I_r\otimes D)$ and ρI_{rn} terms are diagonal, hence $O(nr)$).

4.2 The diagonal preconditioner

Because the transformed regularizer is diagonal, the natural Jacobi preconditioner is *exact and cheap.* Define

$$\bar D := \operatorname{diag}\big(\operatorname{diag}(\bar F^\top \bar F)\big) + \lambda(I_r\otimes D) + \rho I_{rn},$$

whose diagonal $\bar d = \operatorname{diag}(\bar D)$ is, by Lemma 3,

$$\bar d \;=\; \mathrm{vec}\big((\widehat H * \widehat H)^\top(\widehat Z * \widehat Z)\big) \;+\; \lambda\,(\mathbf{1}_r \otimes \operatorname{diag}(D)) \;+\; \rho\,\mathbf{1}_{rn}. \tag{26}$$

This costs $O(q(r^2+n^2))$ once, and each application $\bar D^{-1}z = z/\bar d$ costs $O(nr)$. Unlike a block-Tikhonov choice, this is the true diagonal of the operator, so it captures the per-coordinate scale of the sampling term exactly rather than approximately. The conditioning of $\bar D^{-1}H'_\rho$ is governed by how far $\bar F^\top \bar F$ is from diagonal; in the well-sampled, well-conditioned-Z regime it is close, and PCG converges in few iterations. We make no concentration claim beyond the standard CG bound $\ell = O(\sqrt{\kappa(\bar D^{-1}H'_\rho)}\,\log(1/\varepsilon))$.

4.3 Algorithm

Algorithm 2 PCG for the transformed RKHS mode-k subproblem

Require: triples $\{(i_t, m_t, \tau_t)\}_{t=1}^{q}$, factors $\{A_i\}_{i\neq k}$, kernel K, regularizer λ, shift ρ, tolerance ε.
Ensure: $W \in \mathbb{R}^{n\times r}$.

Setup:
  Eigendecompose $K = UDU^\top$; set $H \leftarrow UD$. ▷ $O(n^3)$, once
  Sampled Khatri–Rao rows $\widehat{Z}(t,:) \leftarrow \bigodot_{i\neq k} A_i(\mathrm{idx}_i(m_t),:)$. ▷ $O(qrd)$
  Gather $\widehat{H}(t,:) \leftarrow H(i_t,:)$ for $t = 1,\dots,q$. ▷ $O(qn)$
  $B(i,:) \leftarrow \sum_{t:i_t=i} \tau_t \widehat{Z}(t,:)$; RHS $g \leftarrow \mathrm{vec}(DU^\top B)$. ▷ $O(qr + n^2 r)$
  Precompute $\bar{d}$ via (26). ▷ $O(q(r^2+n^2))$
PCG: $w_0 \leftarrow 0$, $r_0 \leftarrow g$, $z_0 \leftarrow r_0/\bar{d}$, $p_0 \leftarrow z_0$.
for $\ell = 0, 1, 2, \dots$ until $\|r_\ell\| \le \varepsilon\|g\|$ **do**
  $a_\ell \leftarrow H'_\rho p_\ell$ via (25). ▷ $O(n^2 r + q(r+n))$
  $\alpha_\ell \leftarrow \langle r_\ell, z_\ell\rangle / \langle p_\ell, a_\ell\rangle$.
  $w_{\ell+1} \leftarrow w_\ell + \alpha_\ell p_\ell$; $r_{\ell+1} \leftarrow r_\ell - \alpha_\ell a_\ell$.
  $z_{\ell+1} \leftarrow r_{\ell+1}/\bar{d}$; $\beta_\ell \leftarrow \langle r_{\ell+1}, z_{\ell+1}\rangle / \langle r_\ell, z_\ell\rangle$.
  $p_{\ell+1} \leftarrow z_{\ell+1} + \beta_\ell p_\ell$.
end for
return $W \leftarrow U \cdot \mathrm{unvec}(w_\ell)$. ▷ $O(n^2 r)$

5. Complexity analysis

Setup (PCG): $O\big(n^3 + qrd + q(r^2+n^2) + n^2 r\big)$. The eigendecomposition of K is $O(n^3)$, performed once before the outer alternating iterations; nothing of size M or N is ever formed.

Per iteration: $O\big(n^2 r + q(r+n)\big)$. The $HX = U(DX)$ products ($O(n^2 r)$), the Lemma 1/2 gather–scatter ($O(q(r+n))$), and the diagonal solve ($O(nr)$).

Iterations: $\ell = O\big(\sqrt{\kappa(\bar{D}^{-1}H'_\rho)}\,\log(1/\varepsilon)\big)$.

Total (PCG):

$$O\big(n^3 + qrd + q(r^2+n^2) + \ell\cdot(n^2 r + q(r+n))\big).$$

Comparison. Direct costs $O(qn^2r^2 + n^3r^3)$. PCG removes the Kronecker r^2 blow-up in the iteration cost (down to $O(n^2 r)$) and avoids the $O(n^3 r^3)$ Cholesky, at the price of ℓ matrix-free products. Memory drops from $O(n^2 r^2)$ (storing H) to $O(n^2 + nr + q(r+n))$ (kernel factors, iterate, sampled rows).

6. Why no object of size M or N appears

1. *Kronecker–vec.* $(Z \otimes K)\,\mathrm{vec}(W) = \mathrm{vec}(KWZ^\top)$, but $KWZ^\top \in \mathbb{R}^{n\times M}$ is never formed; only the q entries selected by Ω are needed, and Lemma 1 produces exactly those from the q sampled rows.

2. *Sampling avoids M.* The forward map yields a length-q vector of inner products; the adjoint (Lemma 2) scatters back to an $n \times r$ matrix. Both are $O(q(r+n))$.

3. *Z is never formed.* Only the q rows $\widehat{Z}(t,:)$ that are used appear, each an r-fold Hadamard product of factor rows at $O(rd)$ — the sampled Khatri–Rao trick.

4. *Diagonalizing K.* The transform $K = UDU^\top$ turns the regularizer into a diagonal, enabling the exact diagonal preconditioner (26) via Lemma 3; the only $O(n^3)$ cost is the one-time eigendecomposition.

5. *K is small.* With $n \ll M$, the $O(n^2 r)$ multiplications and $O(n^3)$ factorization are affordable.

7. Definiteness

H is symmetric PSD: $F^\top F \succeq 0$ and $\lambda(I_r \otimes K) \succeq 0$. When $K \succ 0$, $H \succ 0$. When K is only PSD, the shift $\rho > 0$ (equivalently $K \to K + \eta I$) restores strict positive definiteness so that CG is well-defined; in the transformed coordinates this appears as the ρI_{rn} term and a possibly-shifted diagonal D. The transform is orthogonal, so it preserves the spectrum of the regularized operator and the recovered $W = U\bar{W}$ solves the original system.

8. Summary

	Direct	PCG (transformed)
Setup	$O(qrd + n^2 r)$	$O(n^3 + qrd + q(r^2 + n^2))$
Per iter / solve	$O(qn^2r^2 + n^3r^3)$	$O(n^2 r + q(r + n))$
Iterations	1 (Cholesky)	$\ell = O(\sqrt{\kappa}\,\log \frac{1}{\varepsilon})$
Preconditioner	—	exact diagonal (Lemma 3)
Memory	$O(n^2 r^2)$	$O(n^2 + nr + q(r + n))$
Forms H?	yes	no
Touches M or N?	no	no

Conclusion. Diagonalizing the kernel converts the regularizer into a diagonal, after which the three Kronecker-row identities give matrix-free products and an *exact* diagonal (Jacobi) preconditioner. The result solves the RKHS mode-k subproblem in $O(\ell\,(n^2 r + q(r + n)))$ work per iteration plus $O(n^3)$ one-time setup, without forming the $nr \times nr$ system matrix or any object of size $\Omega(M)$ or $\Omega(N)$.

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